

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: MLA03

COURSE CODE: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO3: *Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3, BL6).*

Assessment Tool Weightage: 25%

Dr G. A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Data Interpretation

Duration: 1hr

1. A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what indicates about the learning performance and policy improvement of the agent.

(5 Marks)

2. An epsilon-greedy agent is tested with different epsilon values such as 0.1, 0.3 and 0.5. The cumulative reward obtained is highest when epsilon is 0.1 and lowest when epsilon is 0.5. Justify and apply the concept of exploration vs exploitation to interpret why the reward changes with epsilon and identify which epsilon value provides better performance.

(5 Marks)

3. Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared based on average rewards obtained over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the performance difference between these algorithms and determine which algorithm converges faster based on the observed results.

(5 Marks)

4. An RL agent is trained using different discount factor values such as 0.2, 0.5 and 0.9. The average reward increases as the discount factor increases. Discuss how the discount factor affects long-term reward optimization and justify which discount factor is most suitable for achieving better learning performance. **(5 Marks)**

5. The success rate of an RL agent increases from 45 percent at 50 episodes to 92 percent at 200 episodes. Discuss the improvement in success rate and interpret whether the agent has learned an optimal policy. Provide reasoning based on the trend of the data. **(5 Marks)**

6. An RL agent is trained in the CartPole environment with different learning rates (0.01, 0.05, 0.1). The agent achieves higher stability at 0.05 compared to the other values. Analyze how learning rate influences convergence and justify which value provides the best trade-off between speed and stability. **(5 Marks)**

7. A Q-learning agent is tested with different exploration strategies: ϵ -greedy, softmax, and random exploration. The cumulative reward is highest with ϵ -greedy. Compare the strategies and defend why ϵ -greedy achieves better performance in this scenario. **(5 Marks)**

8. An RL agent trained in a maze environment initially takes 50 steps to reach the goal, but after 300 episodes it reduces to 15 steps. Interpret the reduction in steps and evaluate whether this indicates convergence to an optimal policy. **(5 Marks)**

9. A SARSA agent is trained with discount factors of 0.4, 0.7, and 0.95. The agent achieves the highest long-term reward with 0.95. Discuss the role of discount factor in balancing immediate vs. future rewards and justify why 0.95 is most effective. **(5 Marks)**

10. An RL agent trained in a stochastic environment shows fluctuating rewards in early episodes but stabilizes after 500 episodes. Examine the reward trend and defend how stability indicates policy improvement and robustness of learning. **(5 Marks)**

Code of Conduct:

I _____ M.Vineela(192472071) _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

1. Increasing Reward from 10 to 45

Theory:

In Reinforcement Learning, an agent learns by interacting with an environment. It observes a state, performs an action, and receives a reward. The objective is to learn a policy that maximizes cumulative reward.

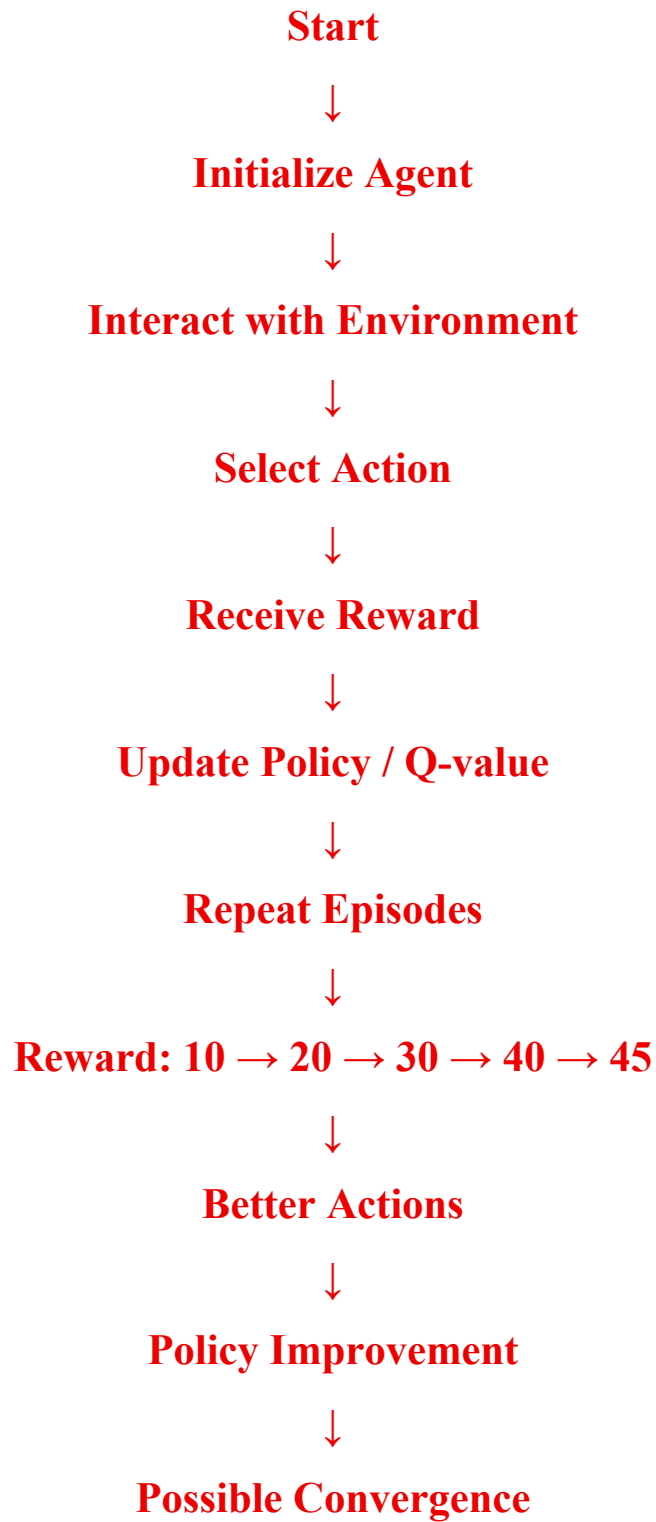
If the reward gradually increases from **10 to 45**, it indicates that the agent is improving through experience. Initially, the agent has limited knowledge and may select poor actions.

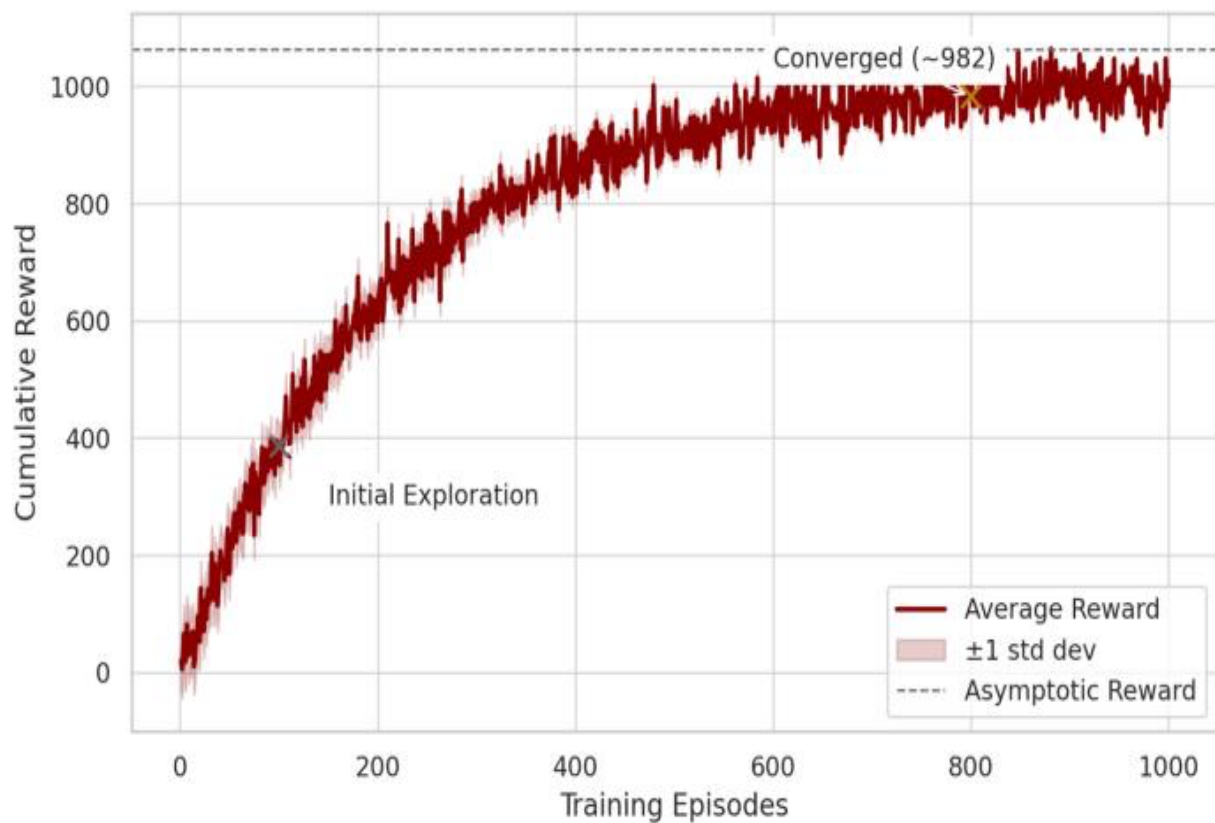
As training continues, it learns which actions produce better rewards and updates its policy or Q-values.

The increasing reward shows:

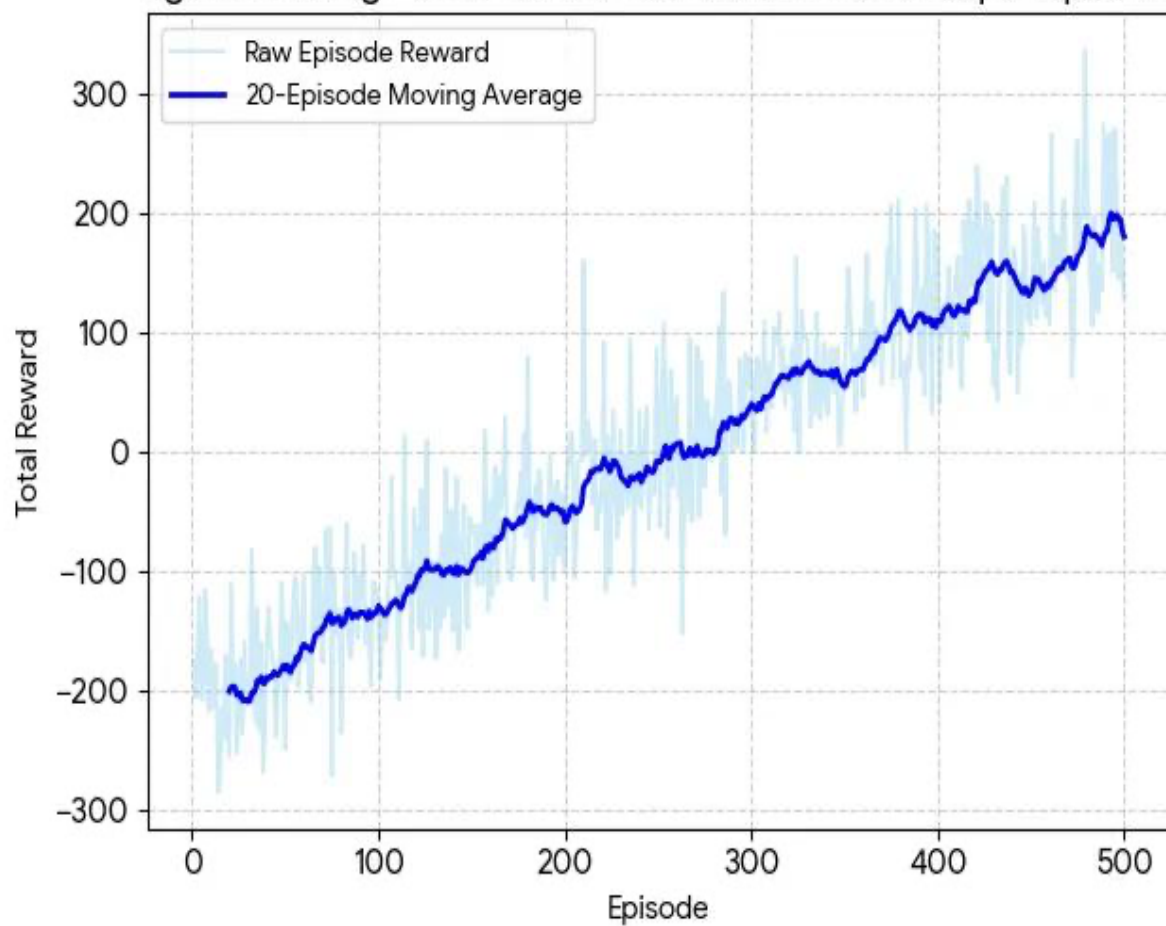
- The agent is learning from experience.
- Action selection is improving.
- The learned policy is becoming more effective.
- The cumulative reward is increasing.
- If the reward becomes stable at a high value, the agent may be approaching convergence.

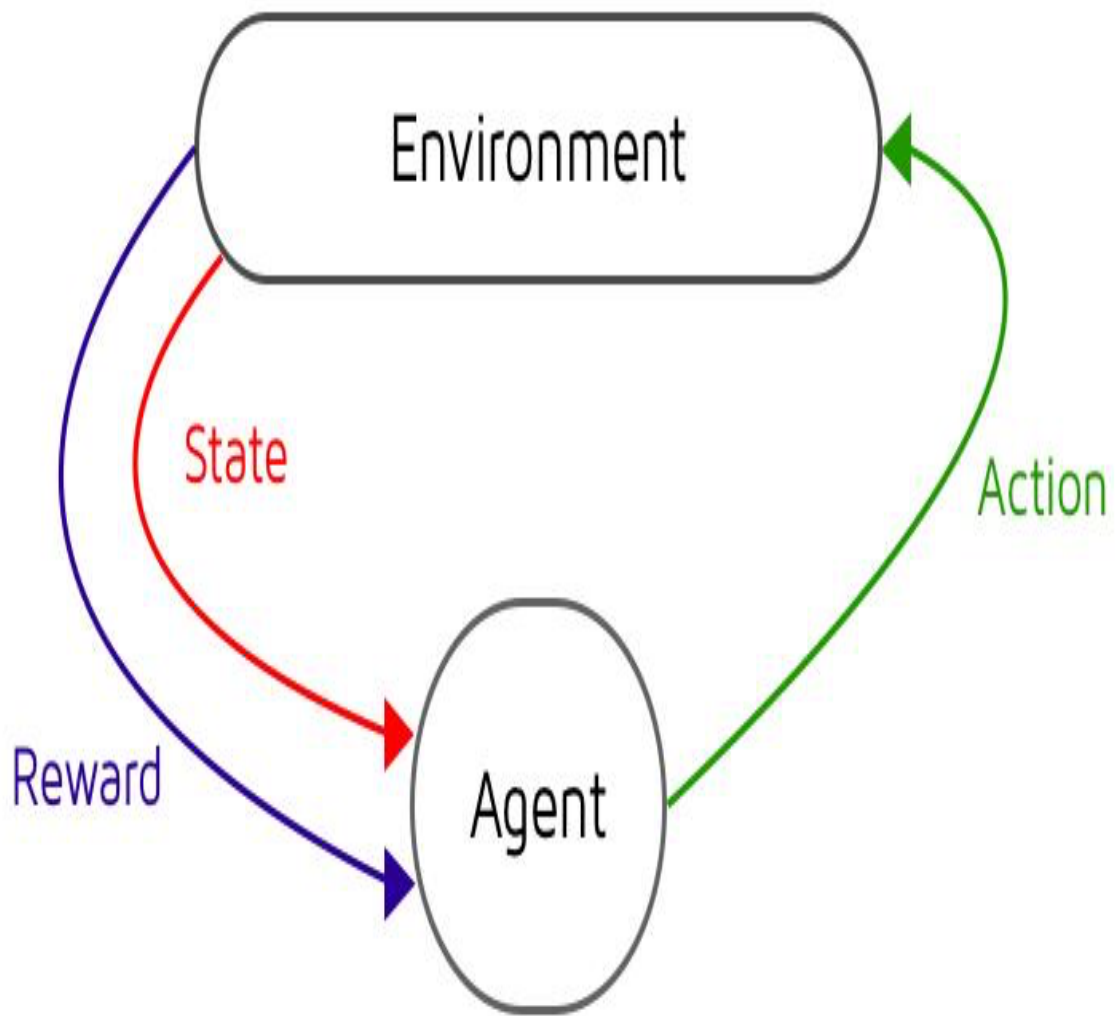
Flow Chart





Agent Training Performance: Cumulative Rewards per Episode





Conclusion

The increase from **10 to 45** indicates successful learning and improvement in the agent's policy. A stable high reward would provide stronger evidence of convergence.

2. Epsilon-Greedy: $\epsilon = 0.1, 0.3$ and 0.5

Theory

The **epsilon-greedy algorithm** balances **exploration** and **exploitation**.

- **Exploration:** Trying new or less-known actions.
- **Exploitation:** Selecting the best-known action.

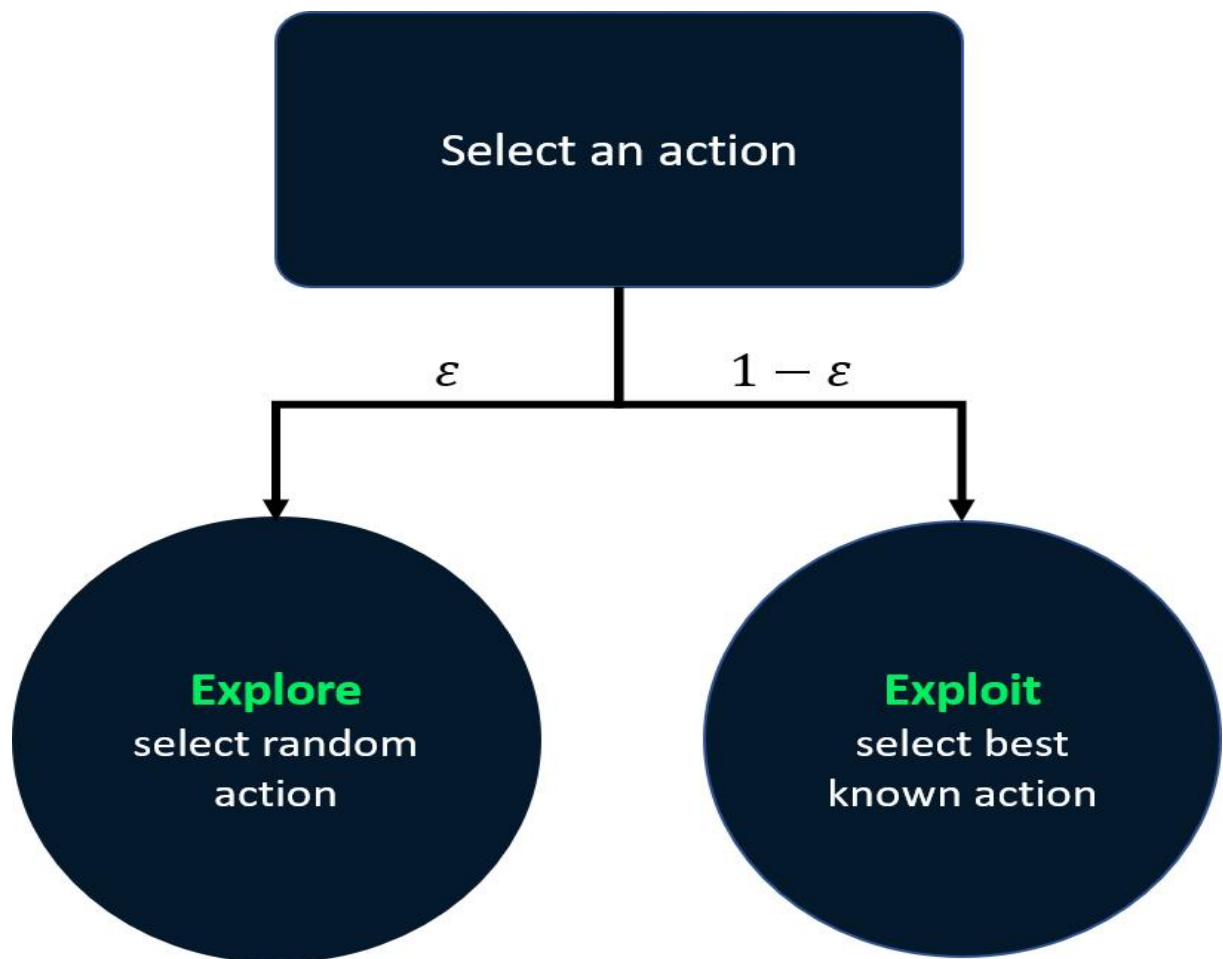
The value of ϵ determines the probability of exploration.

Epsilon	Exploration	Exploitation	Result
$\epsilon = 0.1$	10%	90%	Highest reward
$\epsilon = 0.3$	30%	70%	Medium reward
$\epsilon = 0.5$	50%	50%	Lowest reward

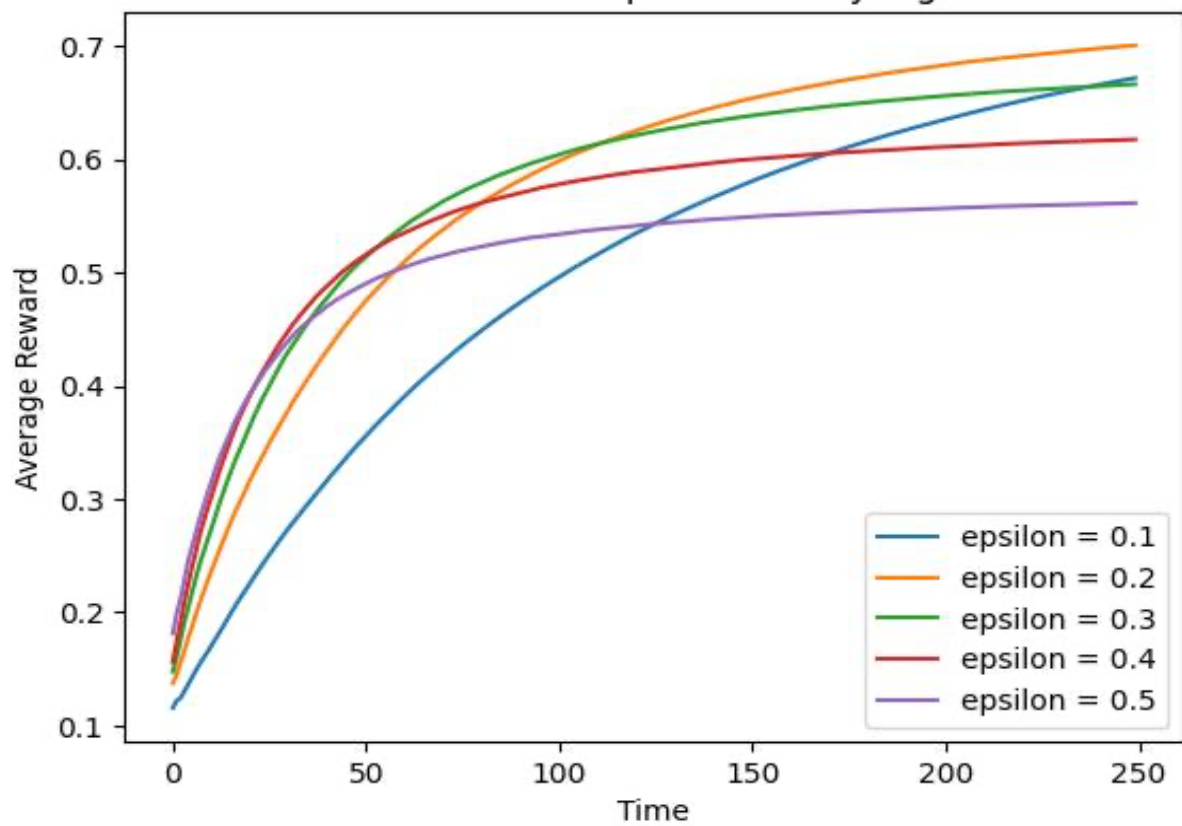
With $\epsilon = 0.1$, the agent mostly exploits its learned knowledge, so it selects rewarding actions more frequently. With $\epsilon = 0.5$, the agent explores more often. This can cause it to select less rewarding actions, reducing cumulative reward. However, exploration is important during early learning because it helps the agent discover better actions. Therefore, the correct epsilon depends on the environment and training stage.

Flow Chart:

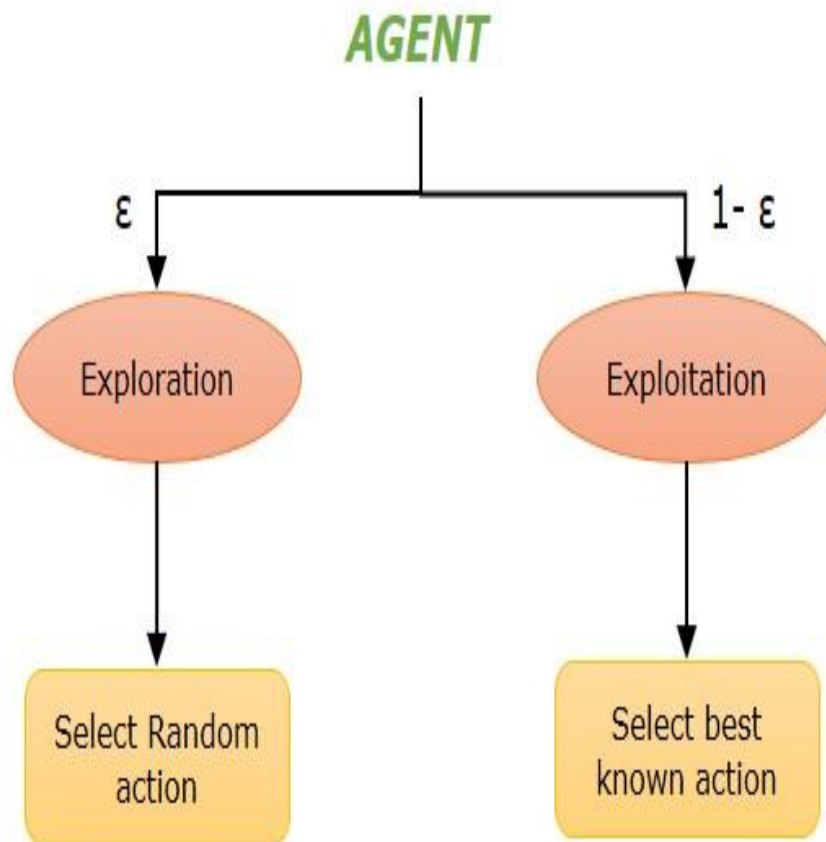




Performance of the Epsilon Greedy Algorithm



Epsilon Greedy Action Selection



Conclusion

$\epsilon = 0.1$ provides the best performance in the given experiment because it provides more exploitation while still allowing some exploration.

3. Q-Learning vs SARSA

Theory

Q-Learning and SARSA are model-free, value-based Reinforcement Learning algorithms.

Q-Learning is an **off-policy** algorithm. It learns using the maximum estimated Q-value of the next state.

SARSA is an **on-policy** algorithm. It learns using the actual next action selected by the current policy.

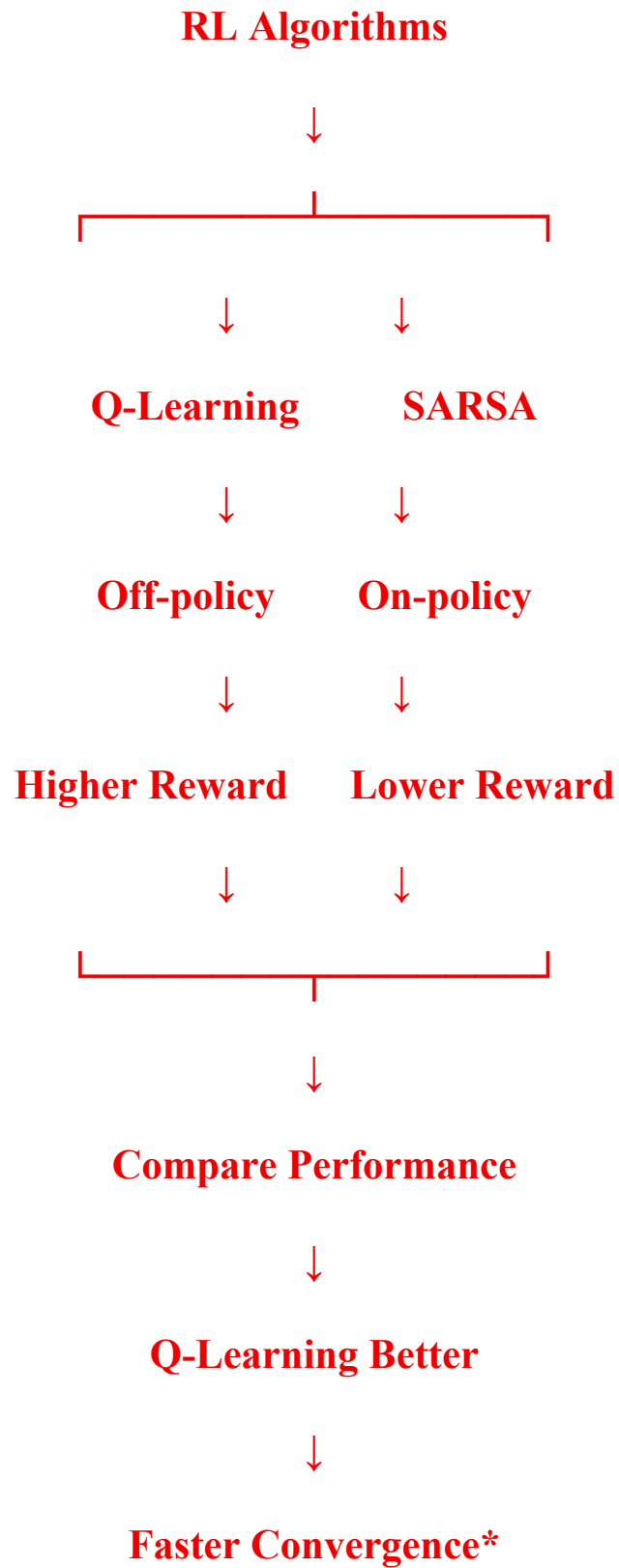
The comparison is:

Feature	Q-Learning	SARSA
Type	Off-policy	On-policy
Learning	Best possible future action	Actual selected action
Observed reward	Slightly higher	Slightly lower
Given performance	Better	Lower

If Q-Learning gives higher average rewards at **100, 200 and 300 episodes**, it indicates better observed performance.

If it reaches a stable high reward earlier, it can also be considered to converge faster in this experiment.

Flow Chart



SARSA vs Q-Learning

ON-POLICY

SARSA



$$Q(s, a) = Q(s, a) + \alpha [r + \gamma Q(s', a) - Q(s, a)]$$

Updates based on the actual action taken

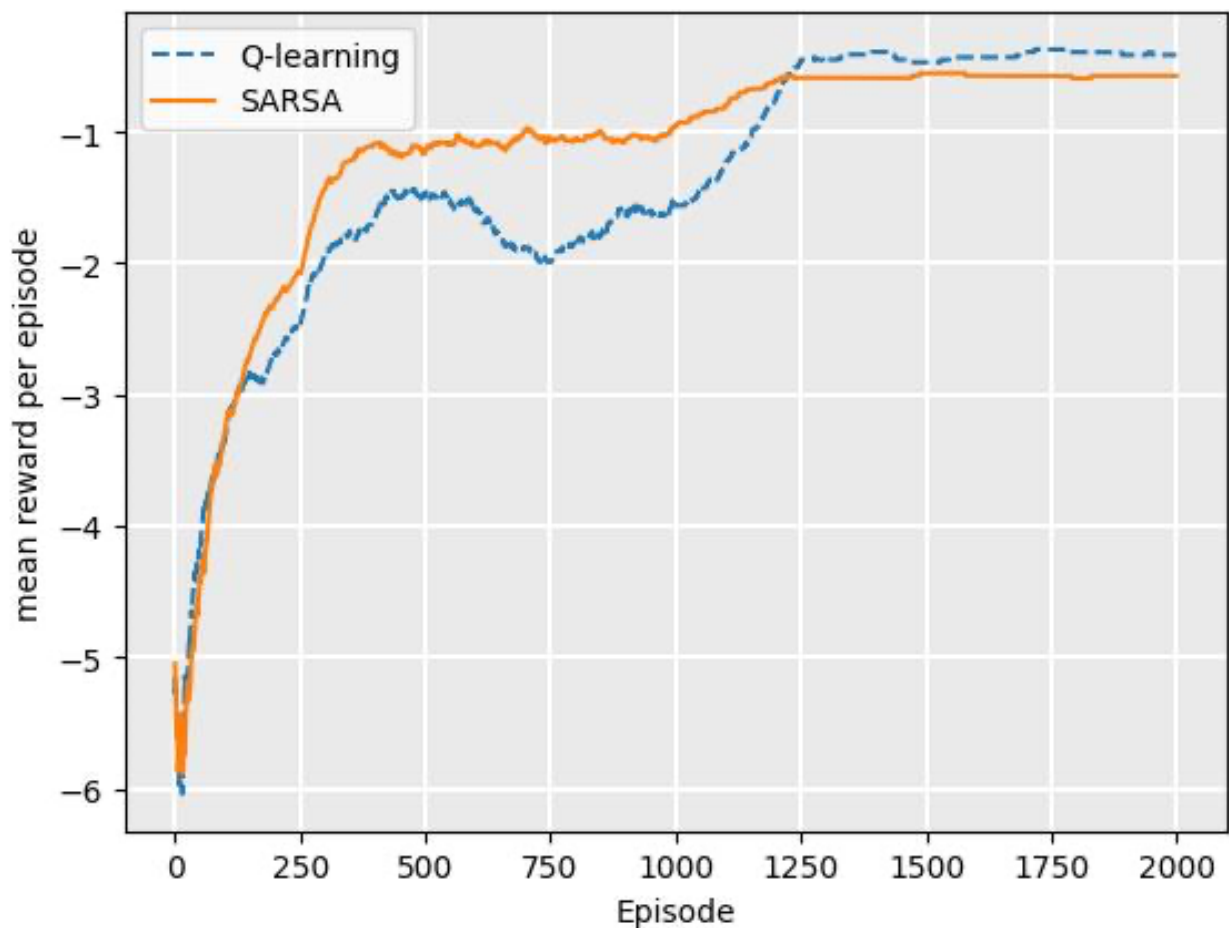
OFF-POLICY

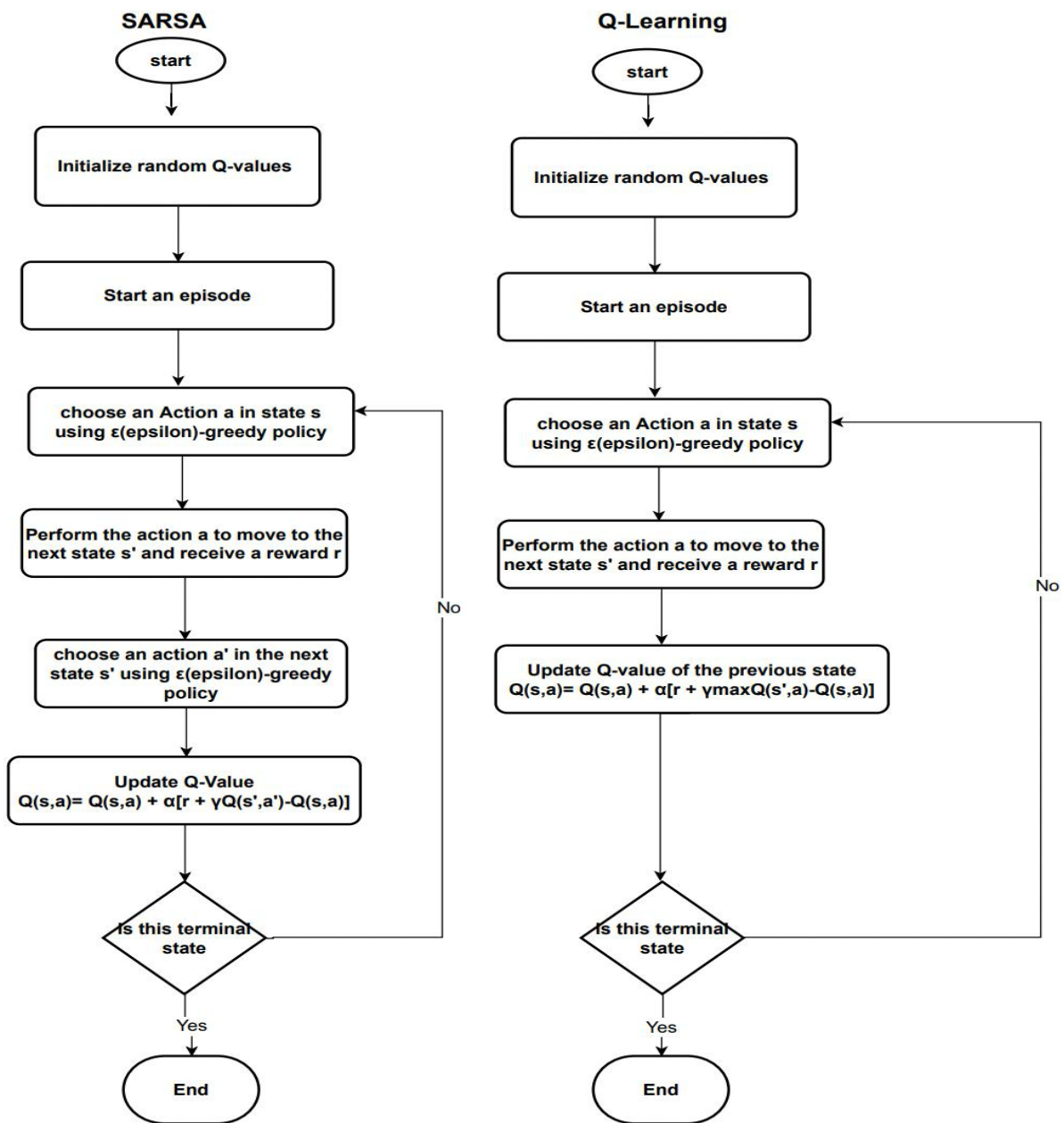
Q-LEARNING



$$Q(s, a) = Q(s, a) + \alpha [r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

Updates based on the optimal next action





Conclusion

Q-Learning consistently produces slightly higher rewards than SARSA. Therefore, Q-Learning shows better observed performance and appears to converge faster in the given experiment.

4. Effect of Discount Factor γ

Theory

The **discount factor** (γ) determines how much importance an agent gives to future rewards.

Its value is generally between **0 and 1**.

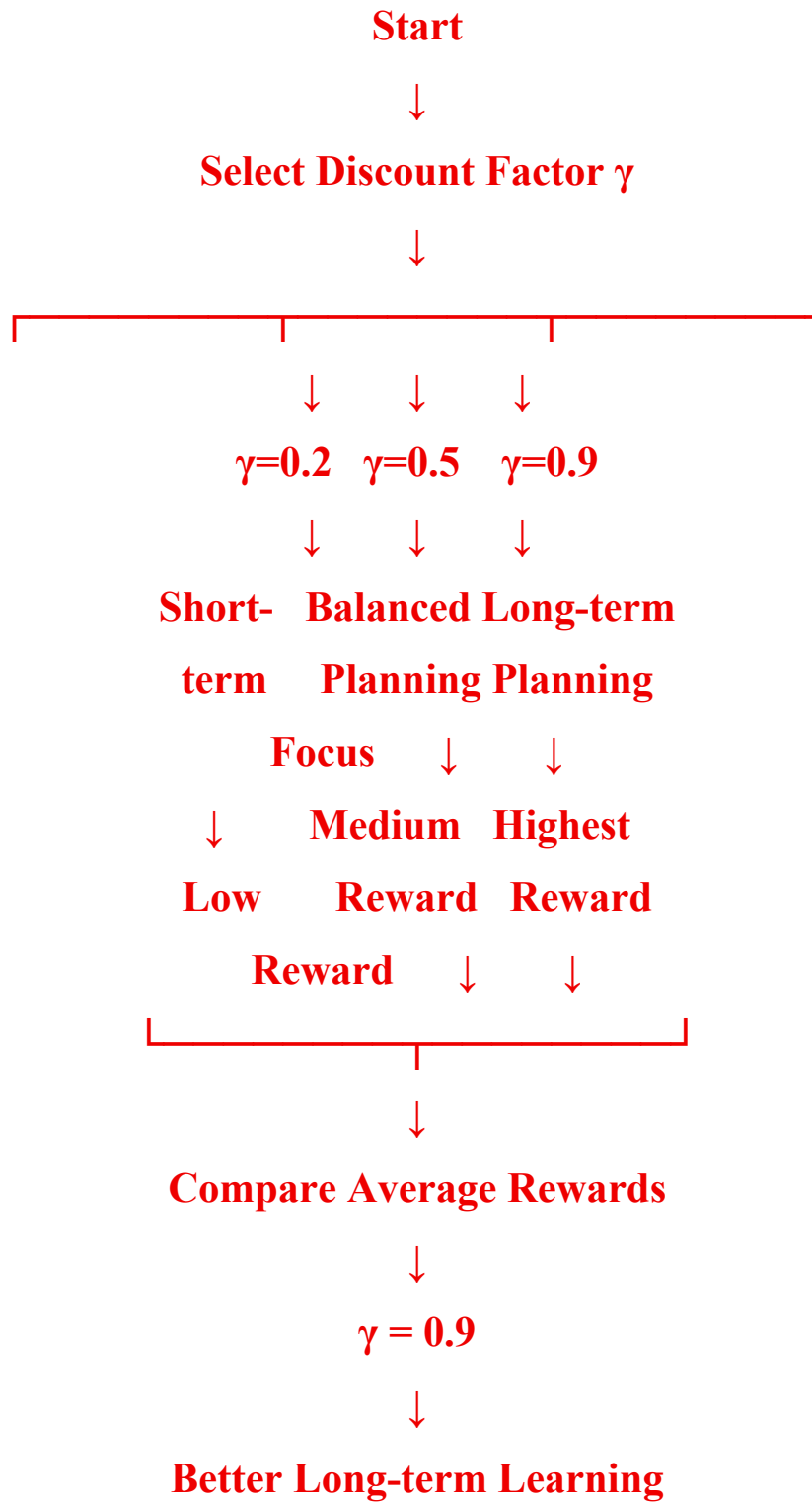
- $\gamma = 0.2$: Strong focus on immediate rewards.
- $\gamma = 0.5$: Moderate consideration of future rewards.
- $\gamma = 0.9$: Strong consideration of future rewards.
- A high discount factor is useful when long-term planning is important.
- The agent may accept a smaller immediate reward if it can obtain a larger reward later.

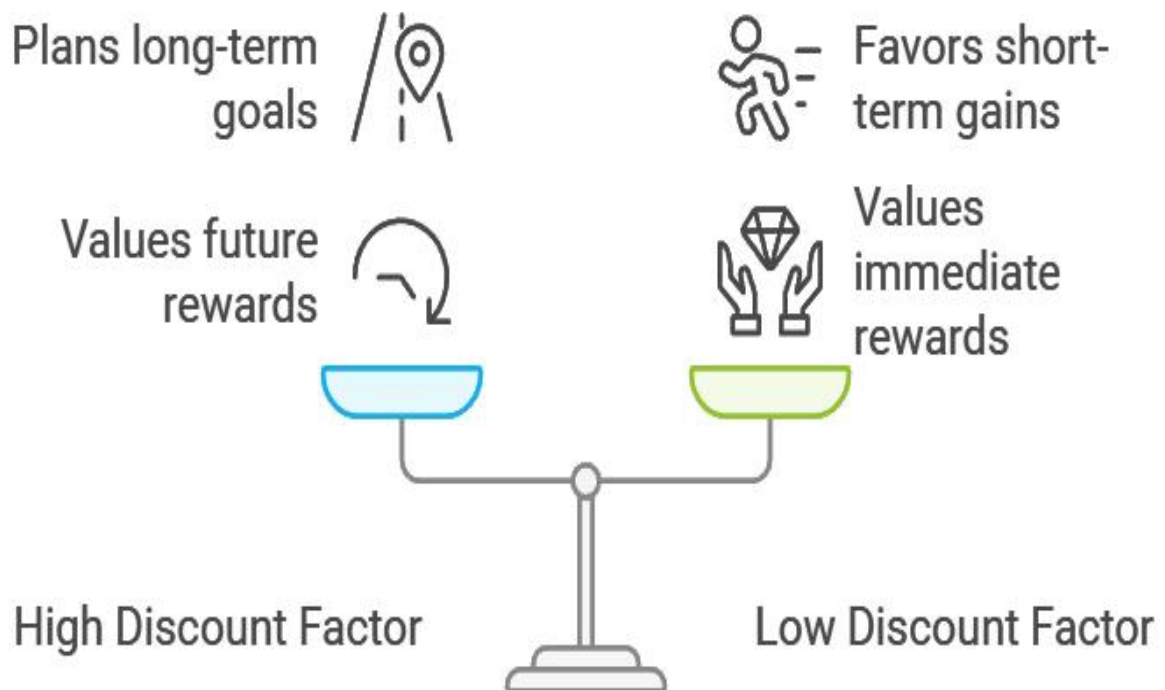
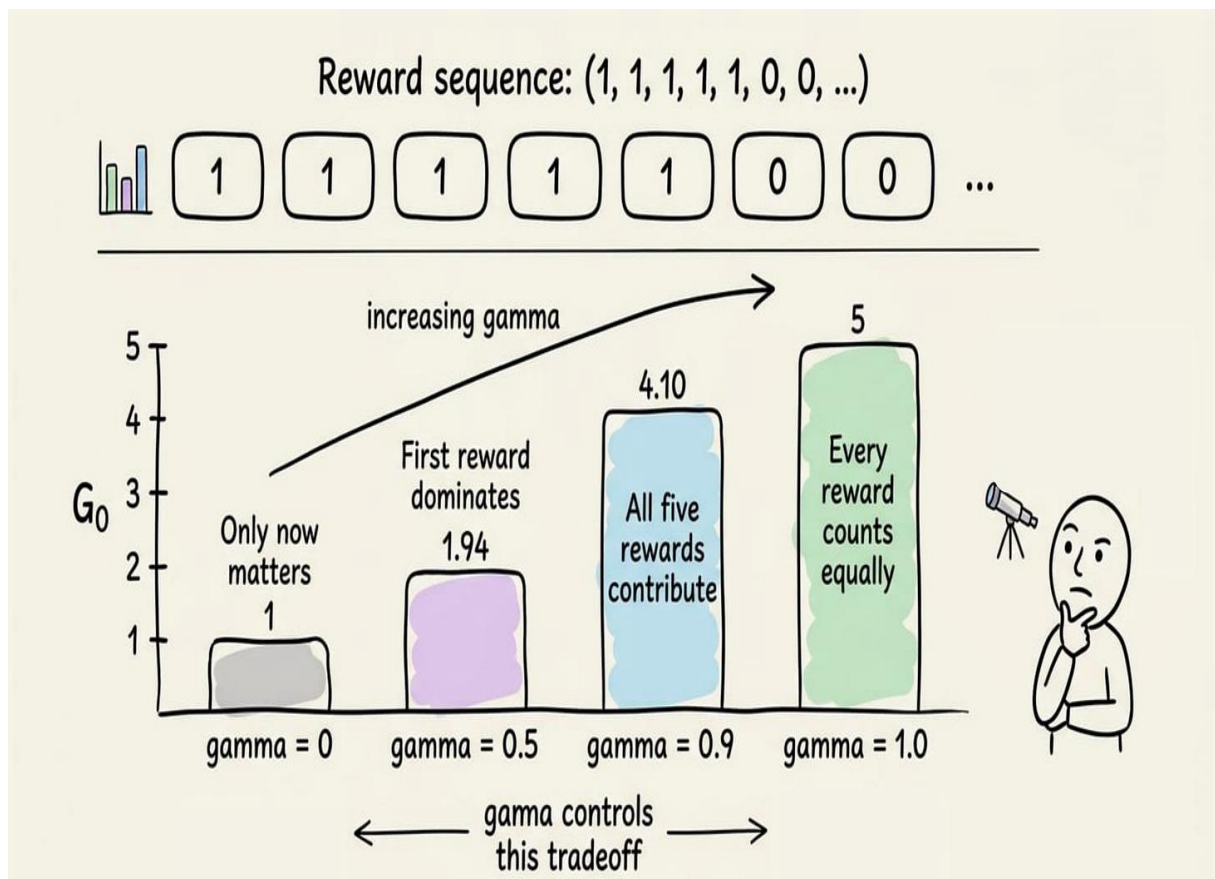
If the average reward increases as:

$$\gamma = 0.2 \rightarrow \gamma = 0.5 \rightarrow \gamma = 0.9$$

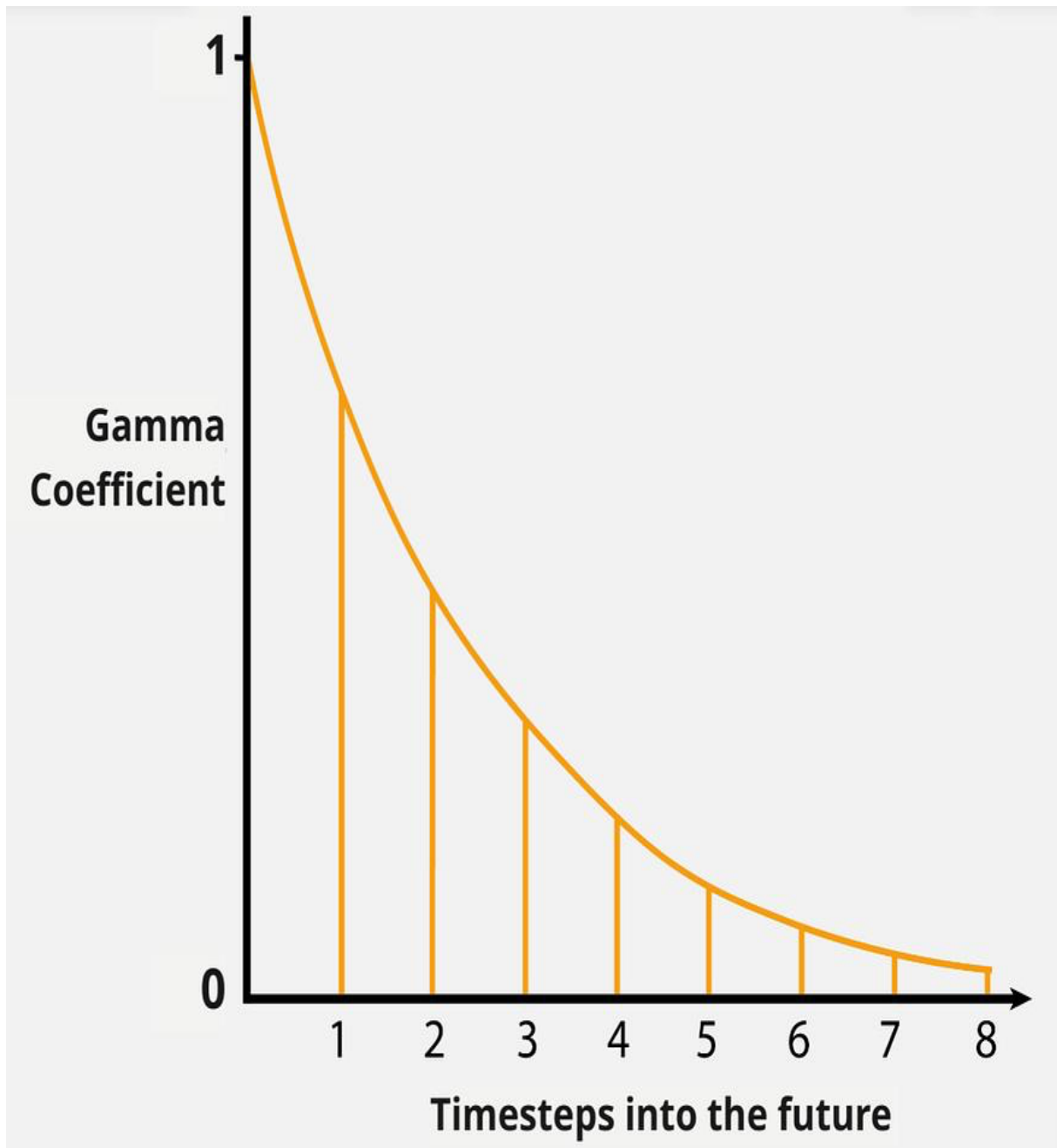
then future rewards are important in the given environment.

Flow Chart





Balancing future and immediate rewards in MDPs.



Conclusion

$\gamma = 0.9$ is the most suitable among the tested values because it produces the highest average reward and allows the agent to consider long-term consequences.

5. Success Rate: 45% to 92%

Theory

The **success rate** represents the percentage of tasks or episodes successfully completed by the RL agent.

The agent's success rate increases from:

45% at 50 episodes → 92% at 200 episodes

The improvement is:

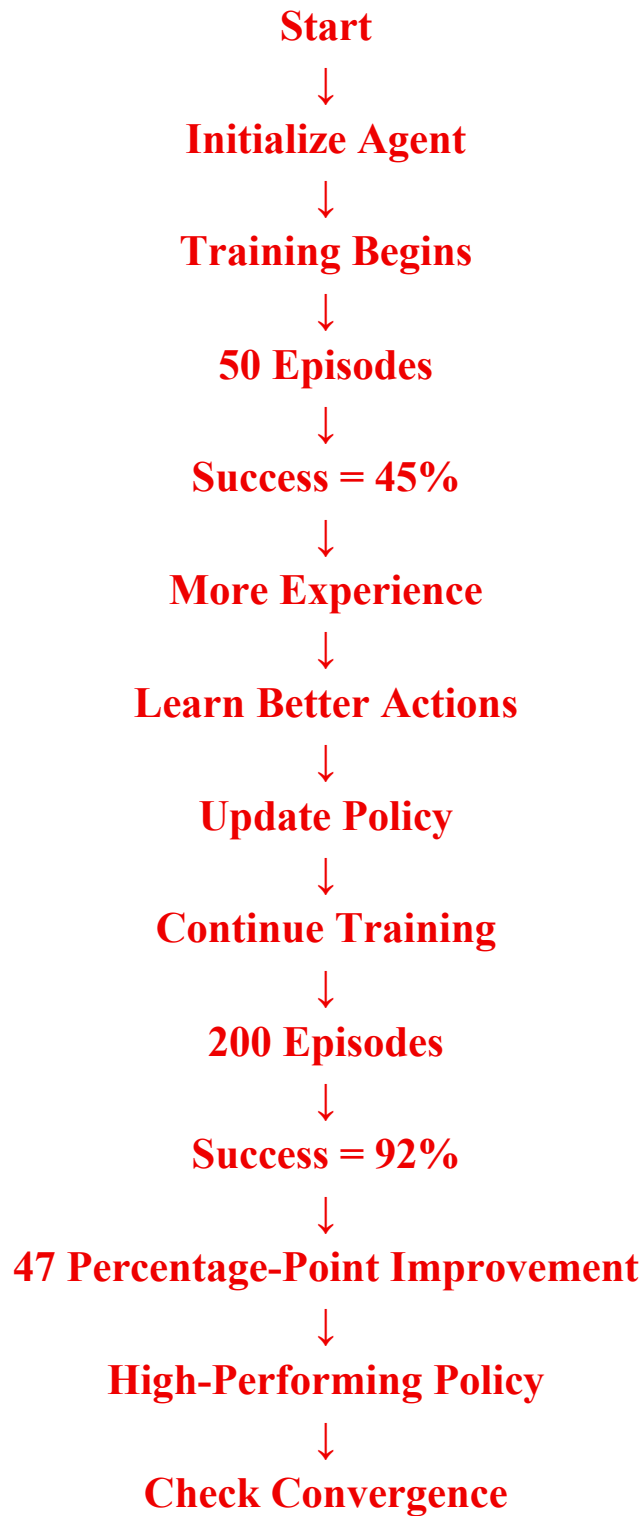
$92 - 45 = 47$ percentage points

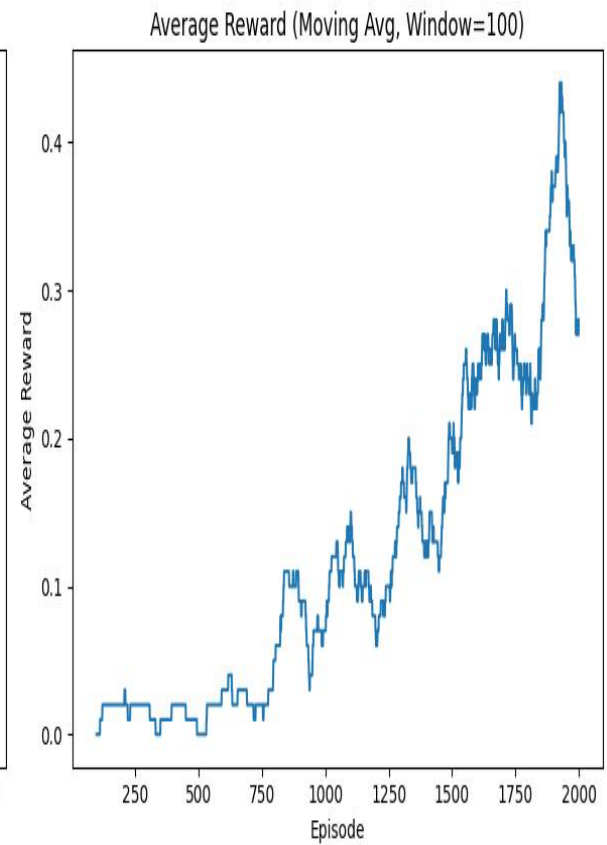
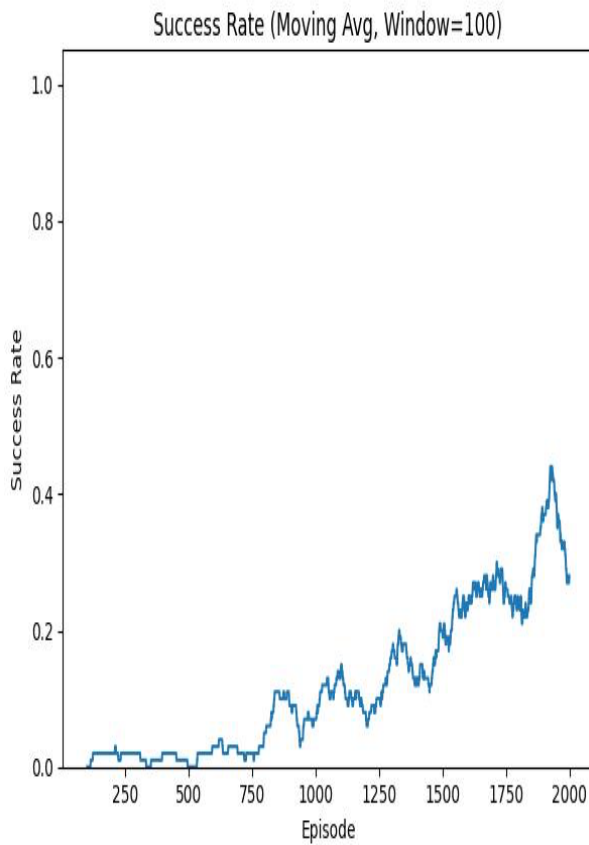
At 50 episodes, the agent has limited experience and makes more mistakes. With continued training, it learns from rewards and penalties, identifies successful actions, and improves its policy.

At 200 episodes, the success rate reaches 92%, showing that the agent has learned a highly effective policy.

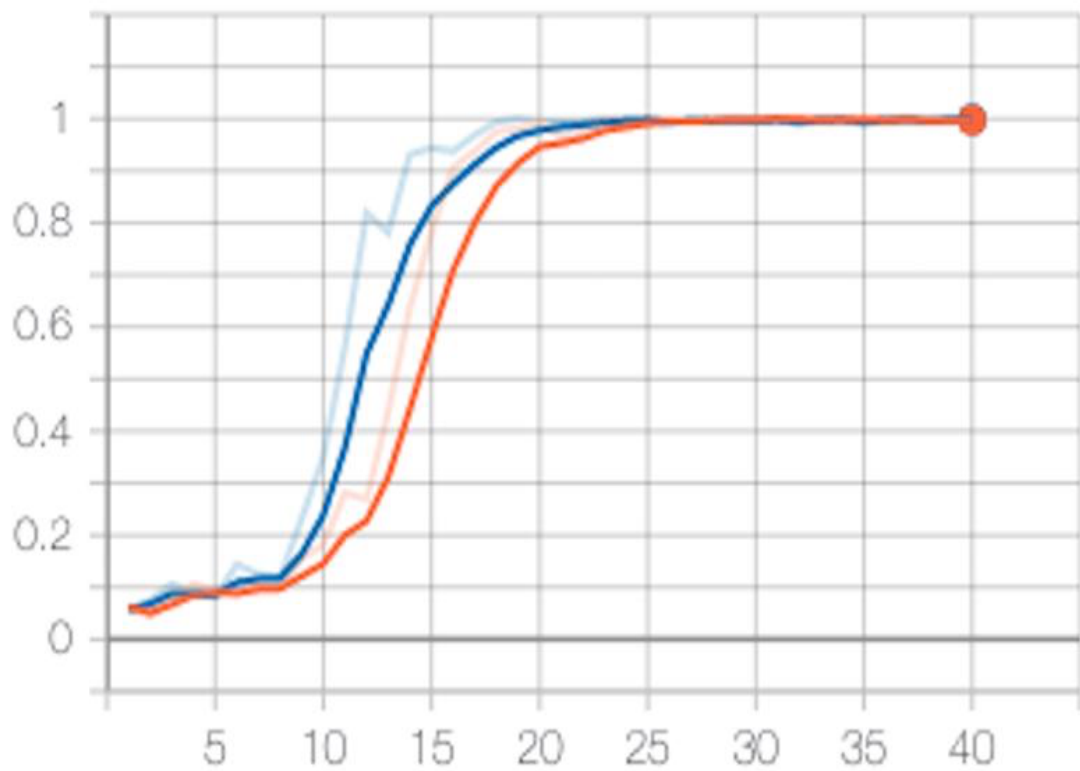
However, a 92% success rate alone does not mathematically prove that the policy is optimal. Further training, stability, convergence, or comparison with an optimal benchmark should be checked.

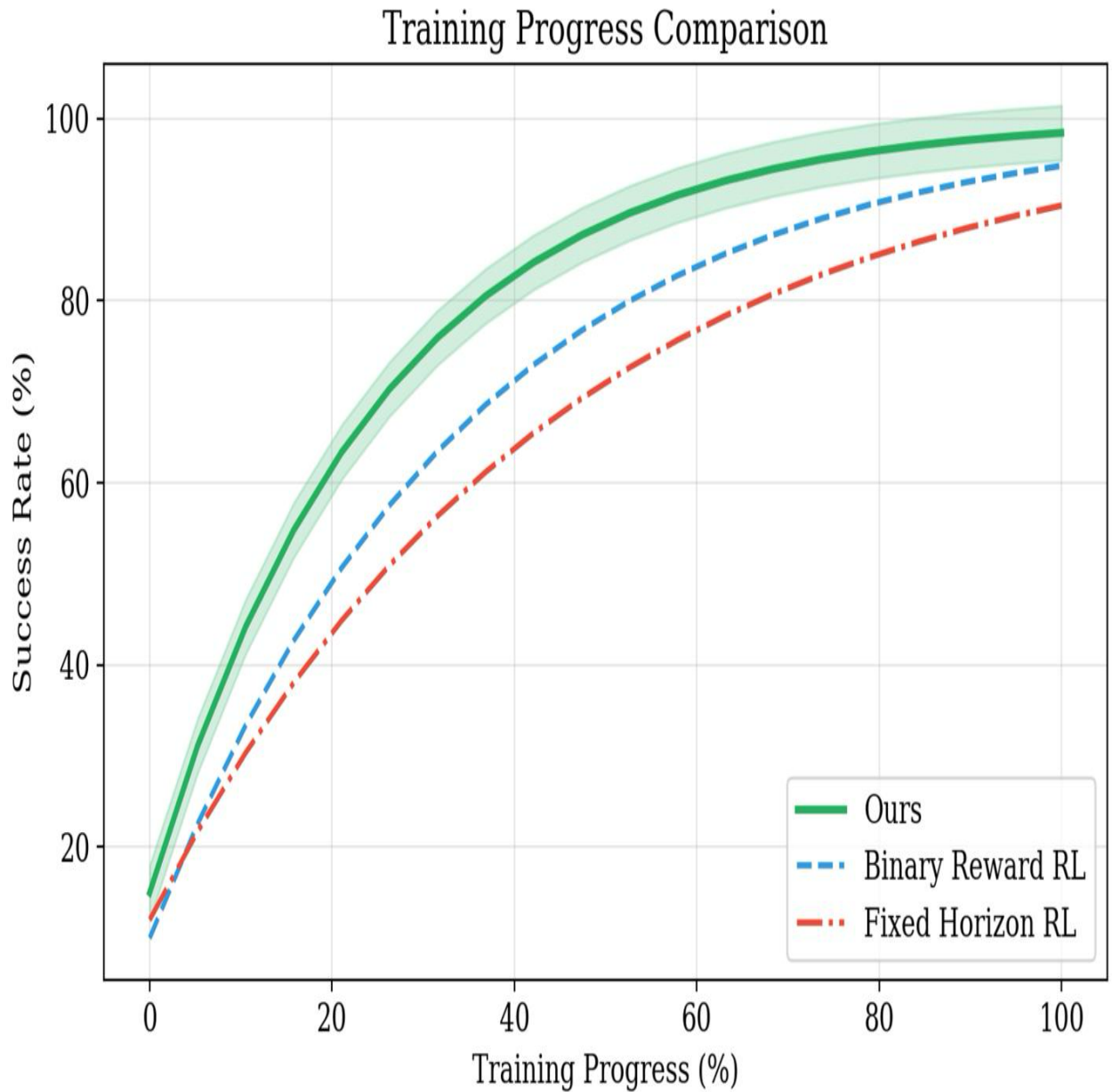
Flow Chart





success_rate
tag: test/success_rate





Conclusion

The increase from 45% to 92% demonstrates significant learning and policy improvement. The agent has learned a high-performing policy. If the 92% success rate remains stable with further training, it gives stronger evidence of policy convergence.

6. Effect of Learning Rate in CartPole

Theory

The **learning rate (α)** controls how much the agent changes its existing knowledge when it receives new information.

The given learning rates are:

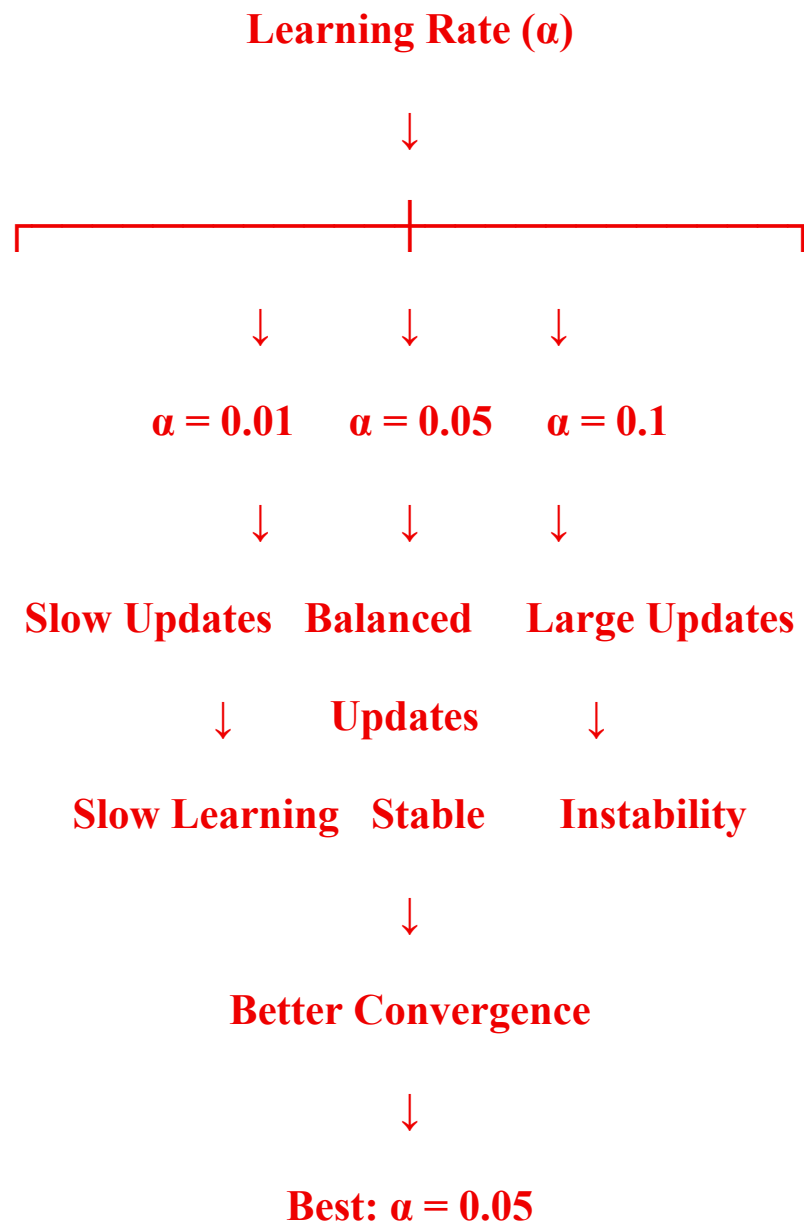
- **$\alpha = 0.01$** : Small updates $\rightarrow$ slow learning and slow convergence.
- **$\alpha = 0.05$** : Moderate updates $\rightarrow$ good balance between learning speed and stability.
- **$\alpha = 0.1$** : Large updates $\rightarrow$ faster learning but may cause instability or fluctuations.

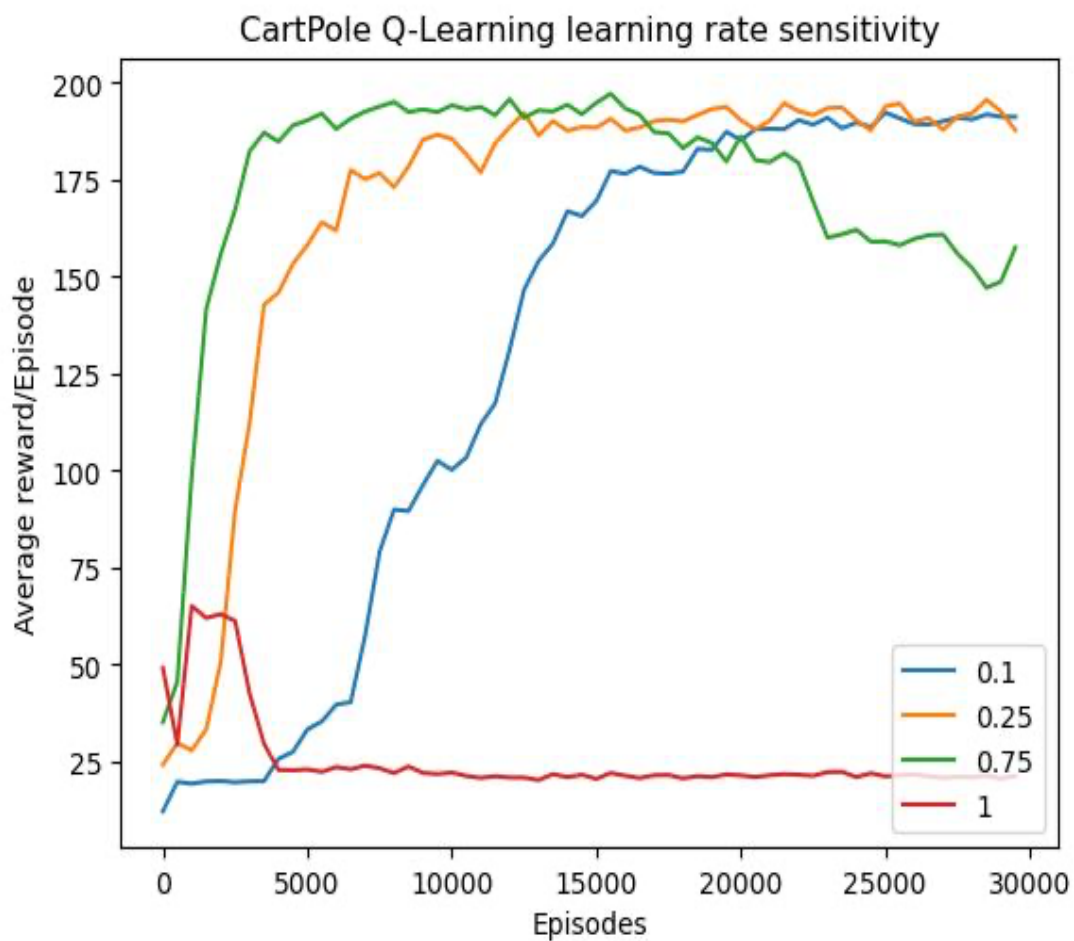
In the CartPole environment, the agent achieves **higher stability at $\alpha = 0.05$** . This means that 0.05 provides a suitable balance between learning speed and stable convergence.

A very small learning rate such as 0.01 may take too many episodes to learn effectively. A very high learning rate such as 0.1 may make the agent change its Q-values too aggressively, causing oscillation or unstable learning.

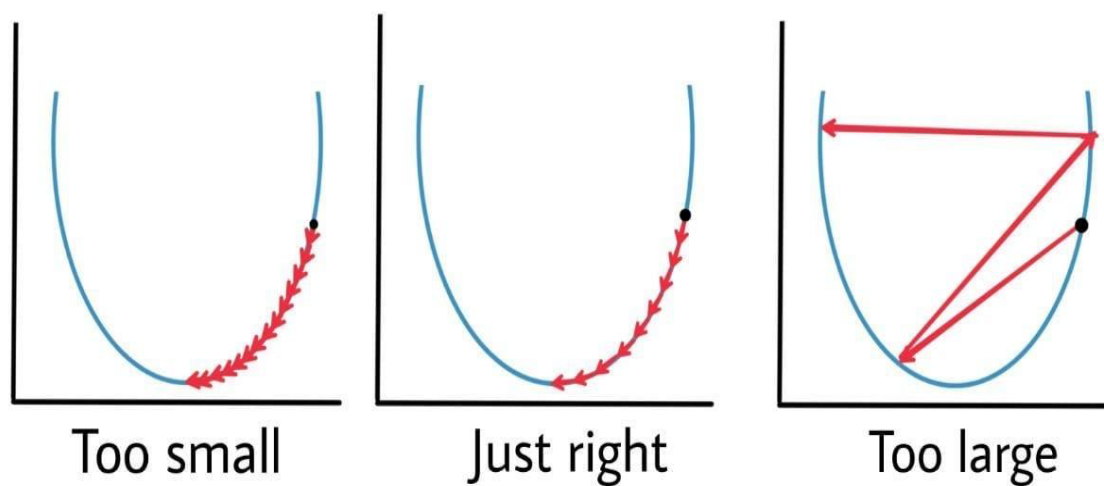
Therefore, **0.05 is the best trade-off** among the tested values.

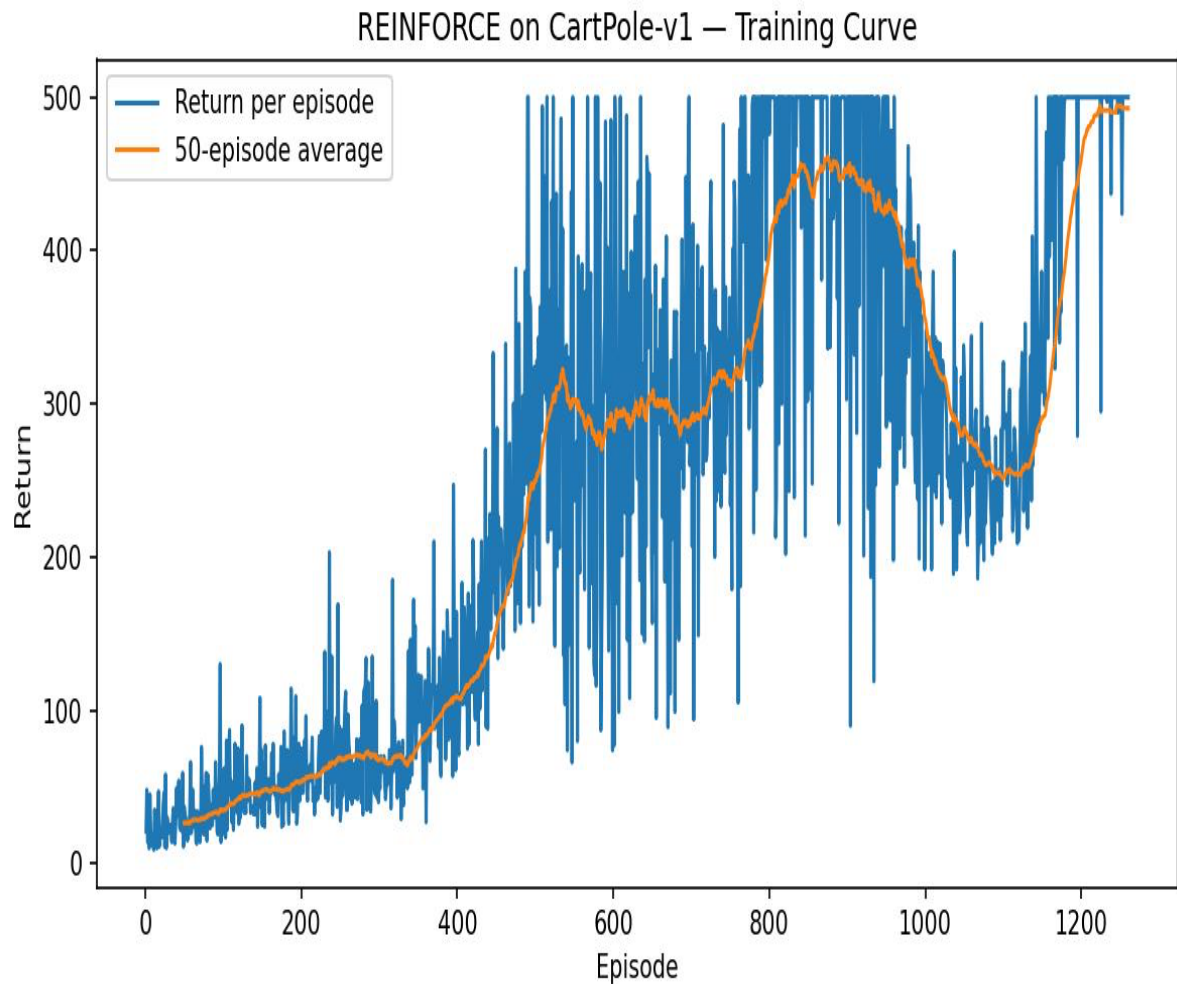
Flow Chart





Learning rate comparison





Conclusion

$\alpha = 0.05$ provides the best performance because it offers a good balance between learning speed and stability. A learning rate that is too low slows convergence, while one that is too high can make learning unstable.

7. Comparison of Exploration Strategies

Theory

Exploration strategies determine how an RL agent chooses actions while learning. The three strategies are:

ϵ -Greedy

- ϵ -greedy selects the best-known action most of the time and randomly explores other actions with probability ϵ .

It provides a simple balance between exploration and exploitation.

Softmax

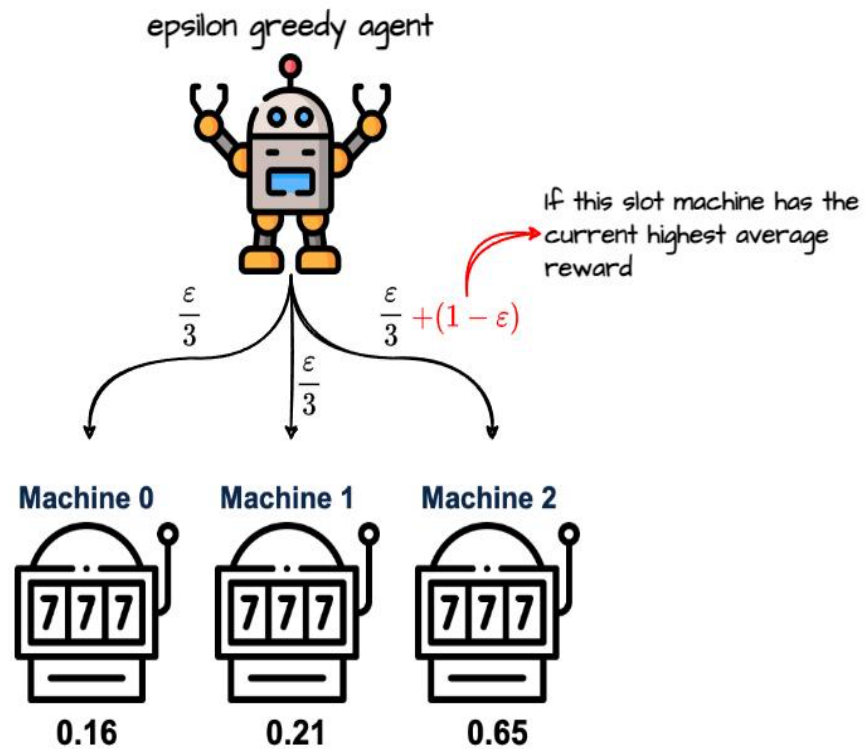
- Softmax assigns a probability to each action based on its estimated value. Actions with higher Q-values have a higher probability of being selected, while lower-valued actions still have some chance.

Comparison:

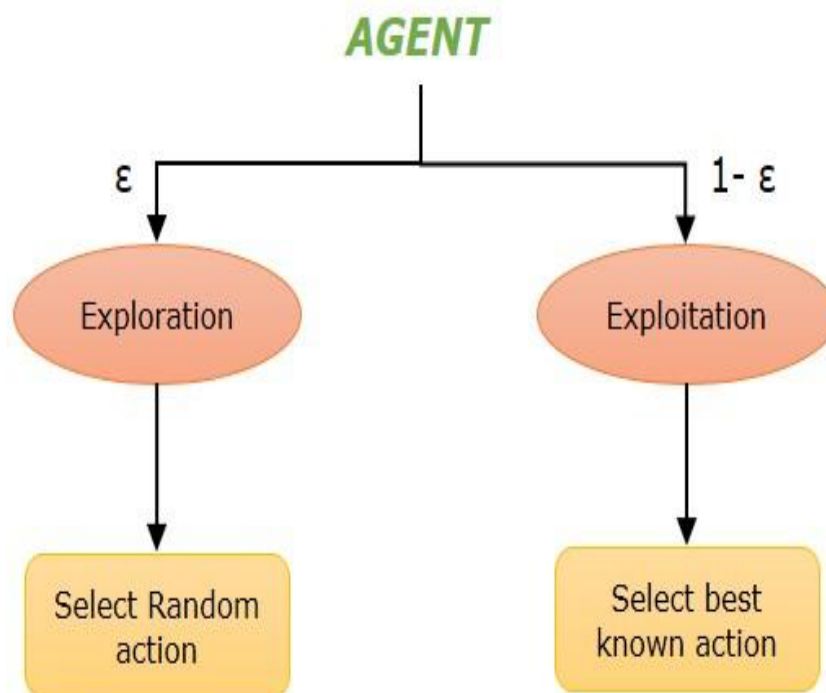
Strategy	Action Selection	Advantage	Performance
ϵ-greedy	Best action + occasional random action	Simple exploration/exploitation balance	Highest
Softmax	Probability based on Q-values	More gradual exploration	Lower
Random	Completely random	Maximum exploration	Lowest

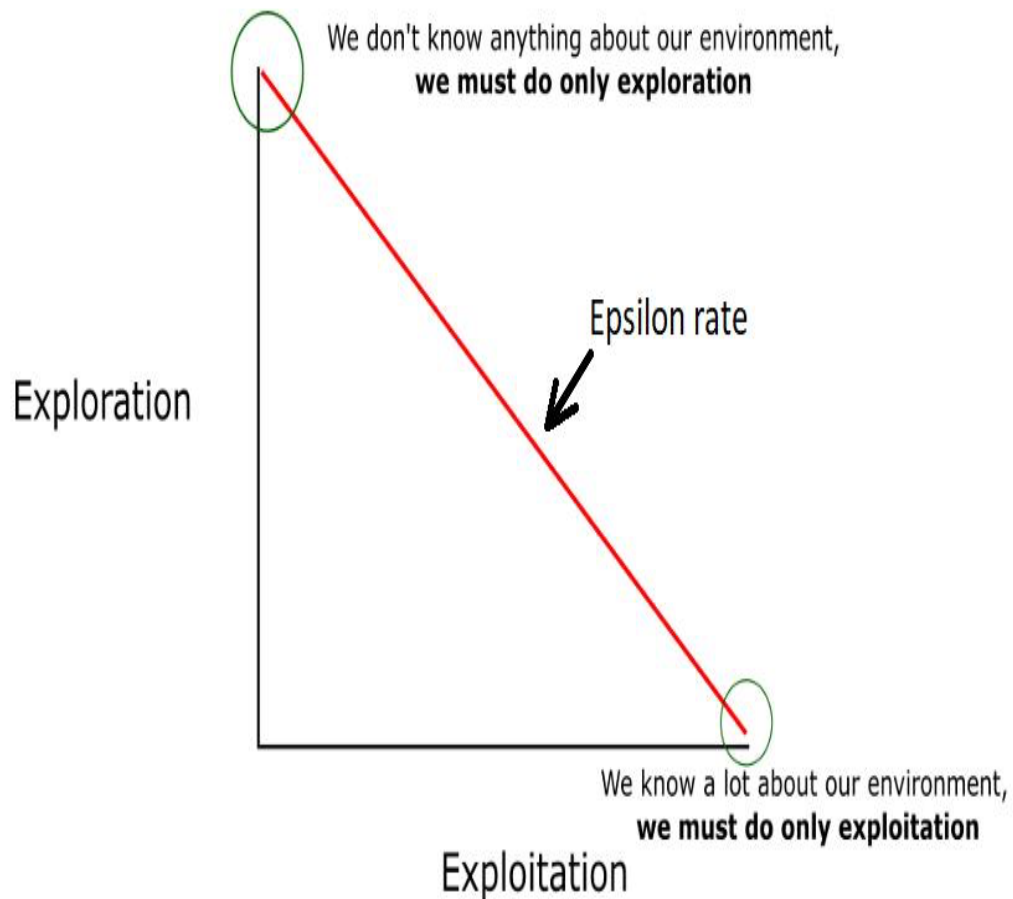
Flow Chart





Epsilon Greedy Action Selection





Conclusion:

ϵ -greedy performs best because it effectively balances exploration and exploitation. It mainly selects the best-known action while still exploring alternative actions. Random exploration wastes many actions, while softmax may explore more than necessary in this particular environment.

8. Reduction in Maze Steps: 50 → 15

Theory

In a maze environment, the number of steps required to reach the goal is an important performance measure. A good RL policy should reach the goal using **fewer steps**.

Initially, the agent takes **50 steps** to reach the goal. This indicates that the agent has limited knowledge of the maze and does not yet know an efficient path.

After training for **300 episodes**, the number of steps decreases to **15**.

The reduction is:

$$50 - 15 = 35 \text{ steps}$$

This represents a significant improvement in efficiency.

As training progresses, the agent learns:

- Which paths lead toward the goal.
- Which paths are inefficient.
- Which actions should be avoided.
- A shorter route to the goal.

Therefore, the reduction from 50 to 15 steps indicates that the policy is improving.

Flow Chart

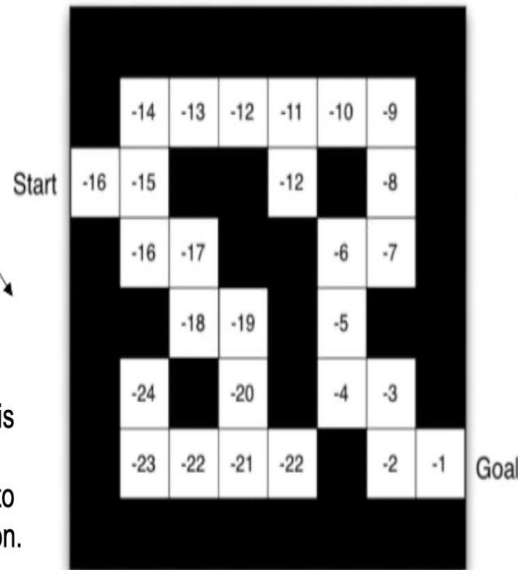


The optimal policy

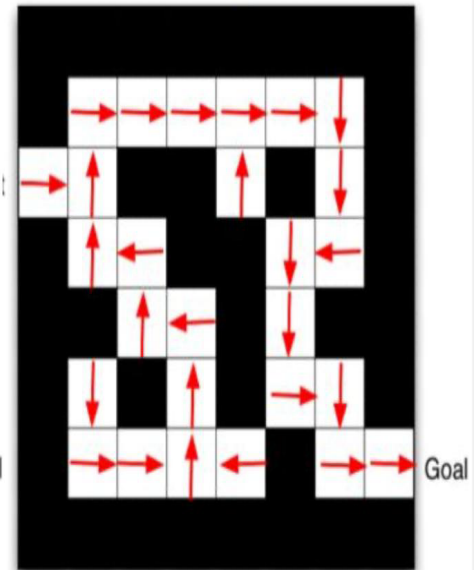
Value of each state (position)

Example:
maze navigation,
-1 reward for each
timestep until maze is
solved.
Common approach to
force quick navigation.

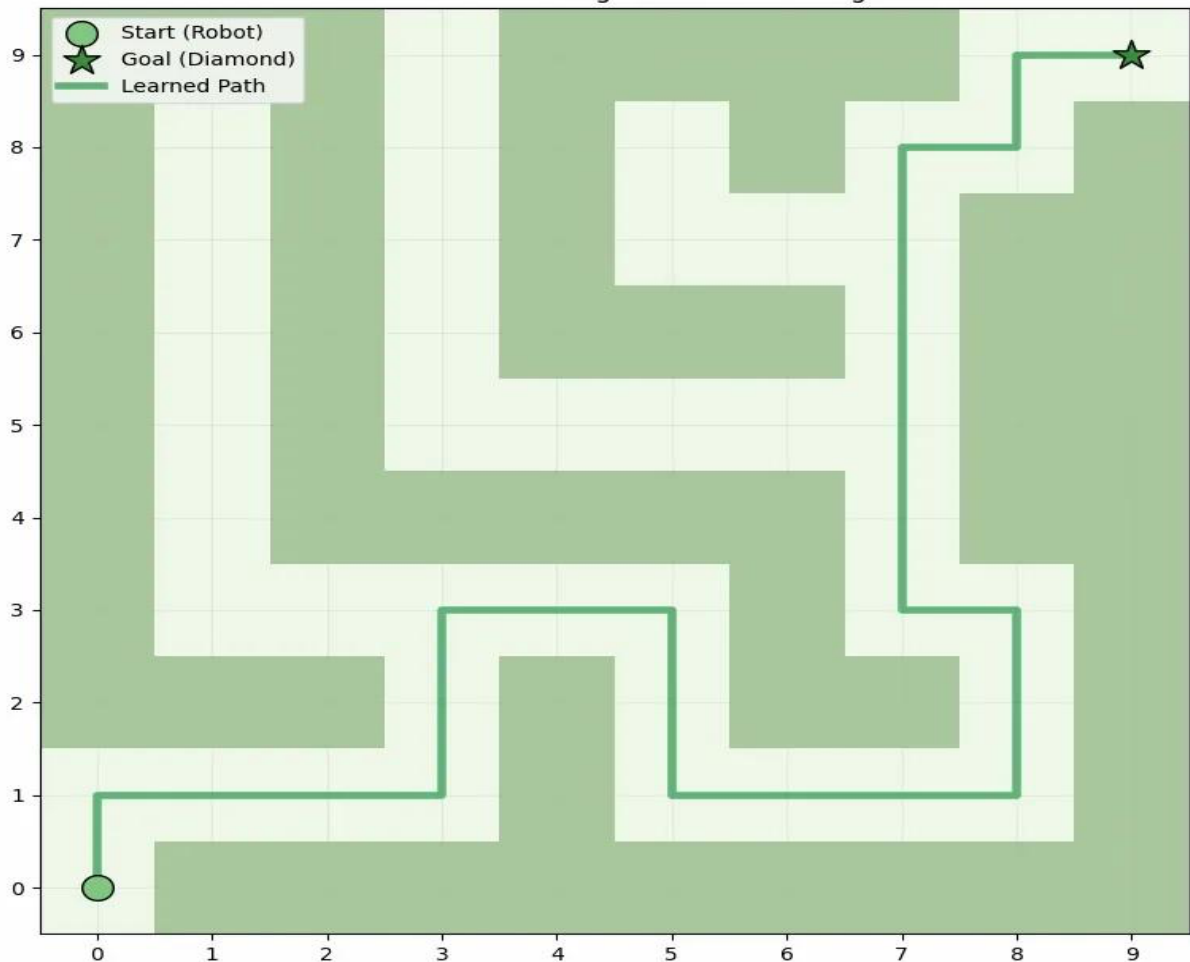
Optimal value function

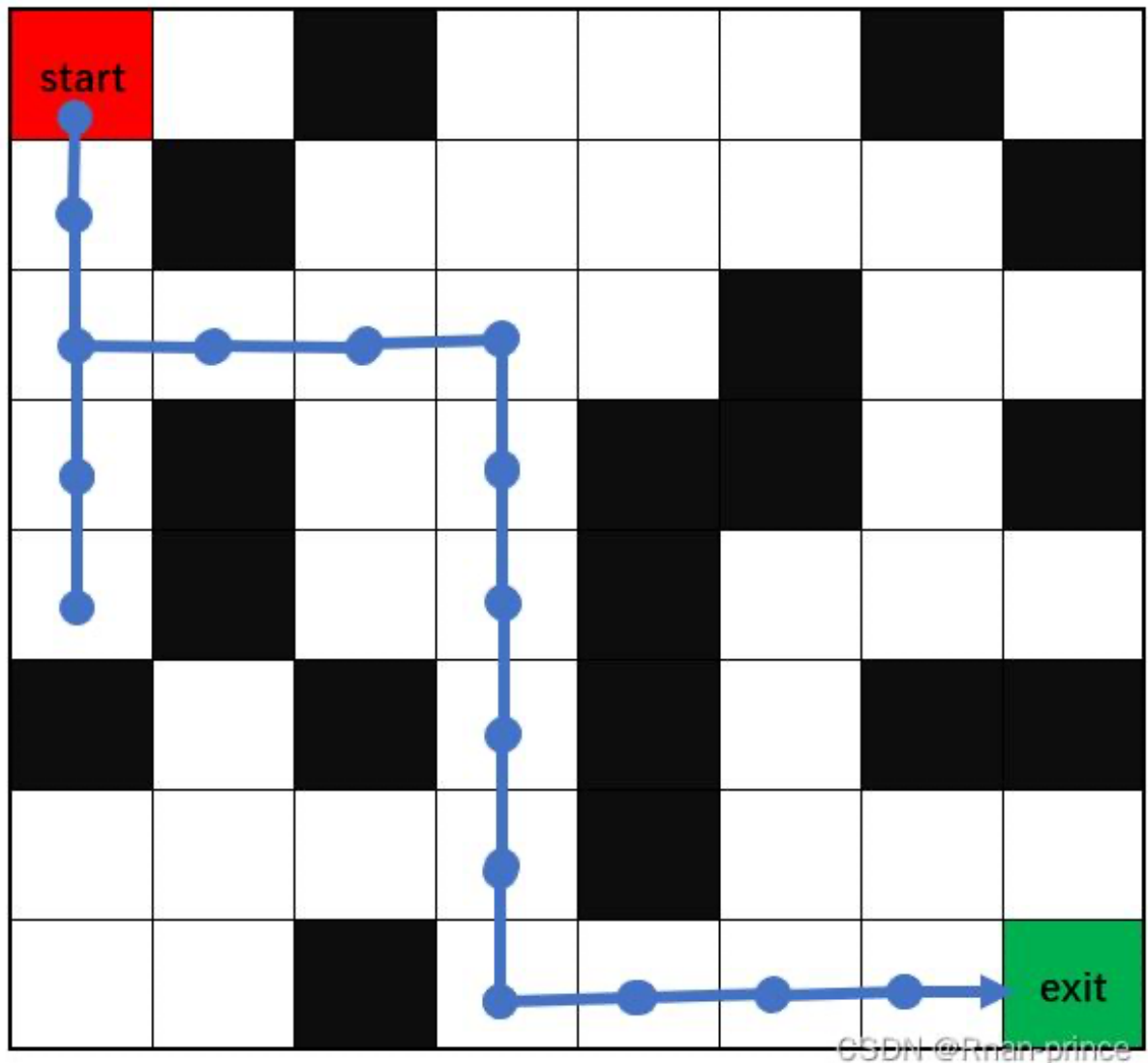


Optimal policy



Reinforcement Learning: Robot Maze Navigation





Conclusion

The reduction from 50 to 15 steps demonstrates significant learning and improved path efficiency. It indicates convergence toward a better policy, but optimality can only be confirmed by comparing 15 steps with the theoretical shortest path.

9. Effect of Discount Factor in SARSA

Theory

The **discount factor** (γ) determines how much importance SARSA gives to future rewards compared with immediate rewards.

The tested values are:

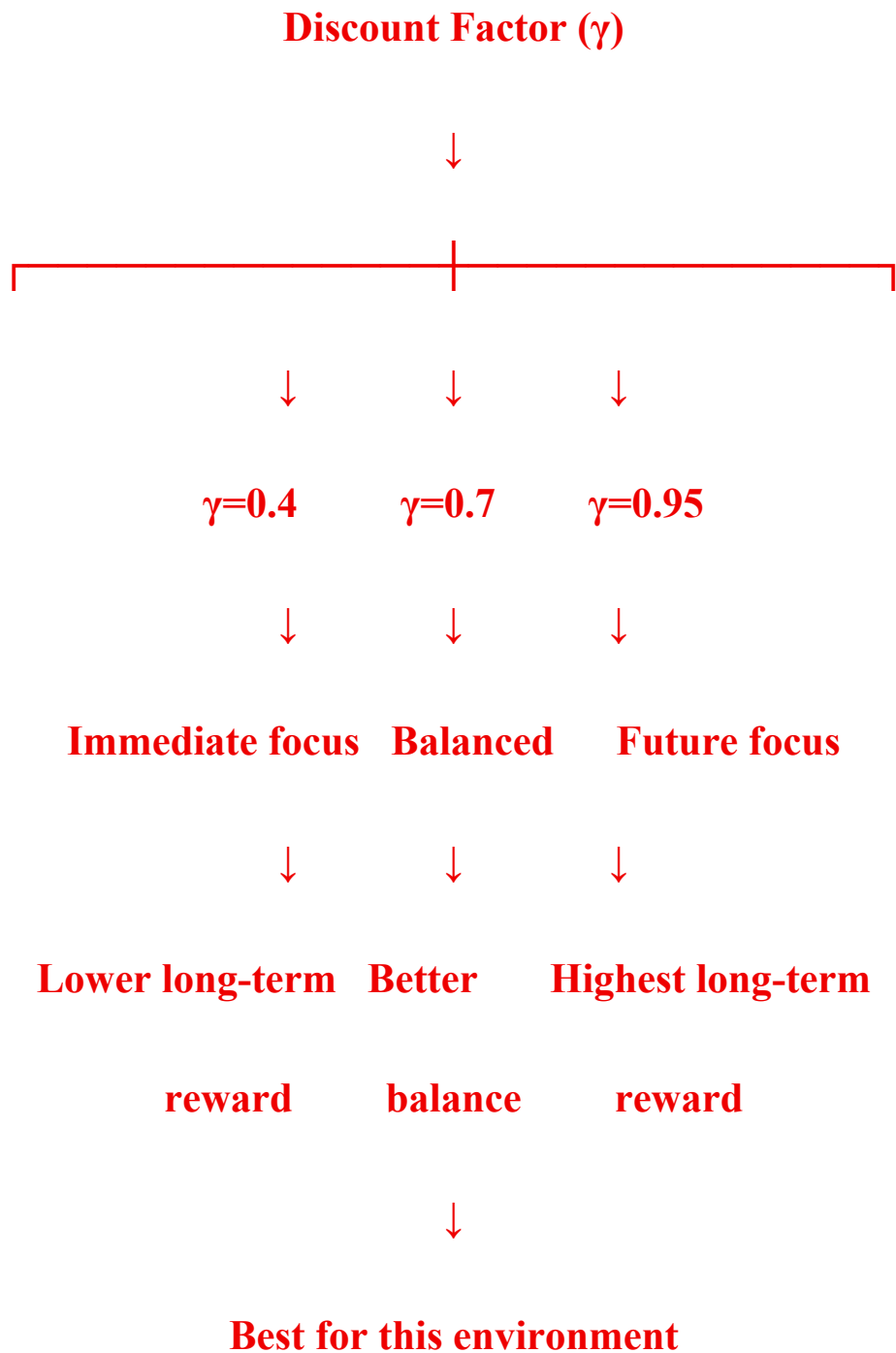
- $\gamma = 0.4$: More focus on immediate rewards.
- $\gamma = 0.7$: Balanced consideration of immediate and future rewards.
- $\gamma = 0.95$: Strong focus on future rewards.

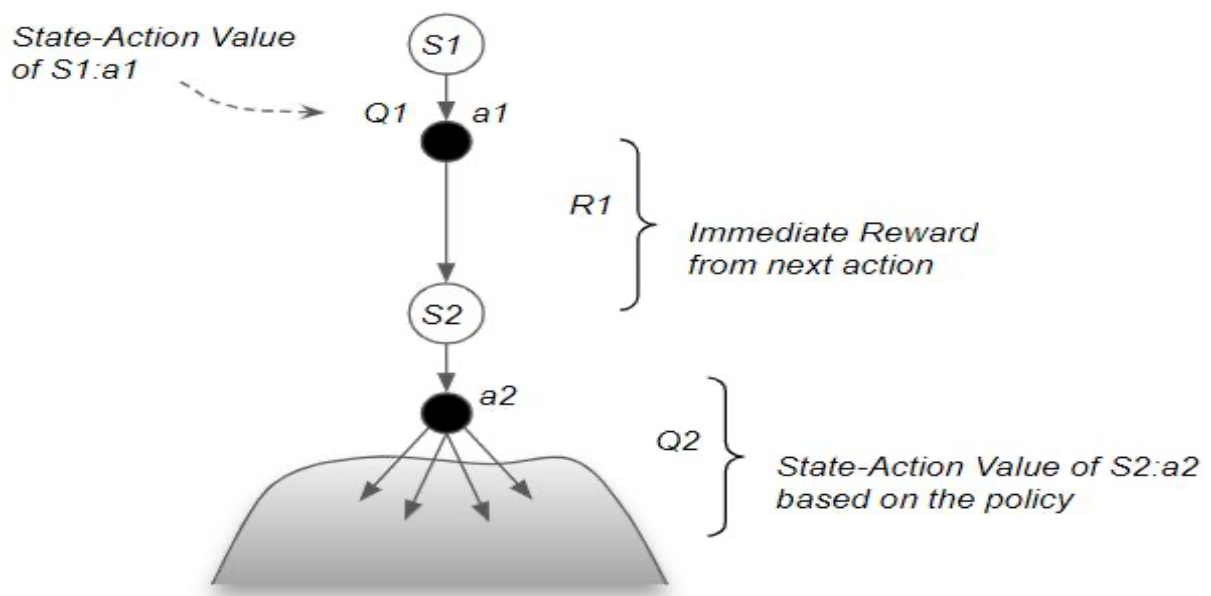
A low discount factor makes the agent behave more like a short-term decision maker. A high discount factor encourages the agent to consider the long-term consequences of its actions.

Since the highest long-term reward is obtained with $\gamma = 0.95$, the environment benefits from long-term planning.

For example, an action may provide a small reward immediately but lead to a much larger reward later. With $\gamma = 0.95$, SARSA gives significant importance to that future reward.

Flow Chart:

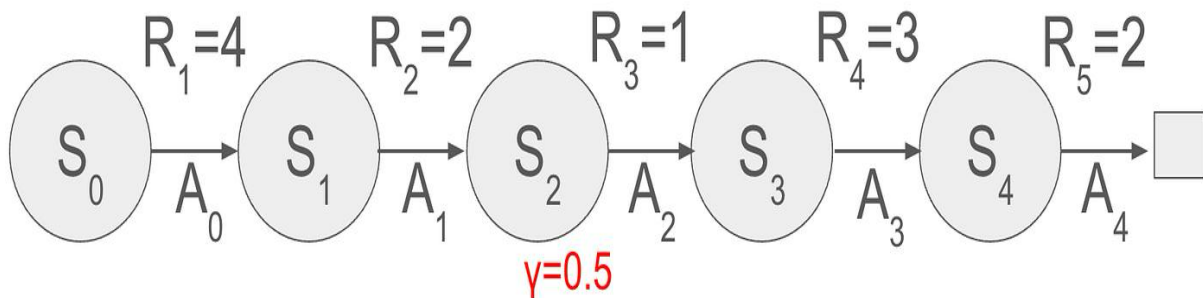




Value of $S1:a1$ = Immediate Reward from next action + Discounted Value of $S2:a2$

$$Q(S_1, a_1) = \mathbb{E}[R_1 + \gamma Q(S_2, a_2)]$$

$$Q(S_t, a_t) = \mathbb{E}[R_t + \gamma Q(S_{t+1}, a_{t+1})]$$



$$G_0 = R_1 + \gamma R_2 + \gamma^2 R_3 + \gamma^3 R_4 + \gamma^4 R_5 = R_1 + \gamma(G_1) = 4 + 0.5 \cdot 3.5 = 5.75$$

$$G_1 = R_2 + \gamma R_3 + \gamma^2 R_4 + \gamma^3 R_5 = R_2 + \gamma(G_2) = 2 + 0.5 \cdot 3 = 3.5$$

$$G_2 = R_3 + \gamma R_4 + \gamma^2 R_5 = R_3 + \gamma(G_3) = 1 + 0.5 \cdot 4 = 3$$

$$G_3 = R_4 + \gamma R_5 = R_4 + \gamma(G_4) = 3 + 0.5 \cdot 2 = 4$$

$$G_4 = R_5 = 2$$



Balancing future and immediate rewards in MDPs.

Conclusion

$\gamma = 0.95$ is the most effective value because it gives strong importance to future rewards and produces the highest long-term reward. This indicates that long-term planning is important in the given environment.

10. Fluctuating Rewards and Stabilization

Theory

In a stochastic environment, the same action may sometimes produce different outcomes because of randomness or uncertainty. Therefore, reward values may fluctuate significantly during the early training episodes.

Initially, the agent has limited knowledge and explores different actions. As a result, rewards may increase and decrease irregularly.

After approximately **500 episodes**, the rewards become more stable.

This stabilization indicates that the agent has learned a more consistent policy. It has identified actions that generally produce good outcomes despite the randomness in the environment.

The reward trend can be interpreted as:

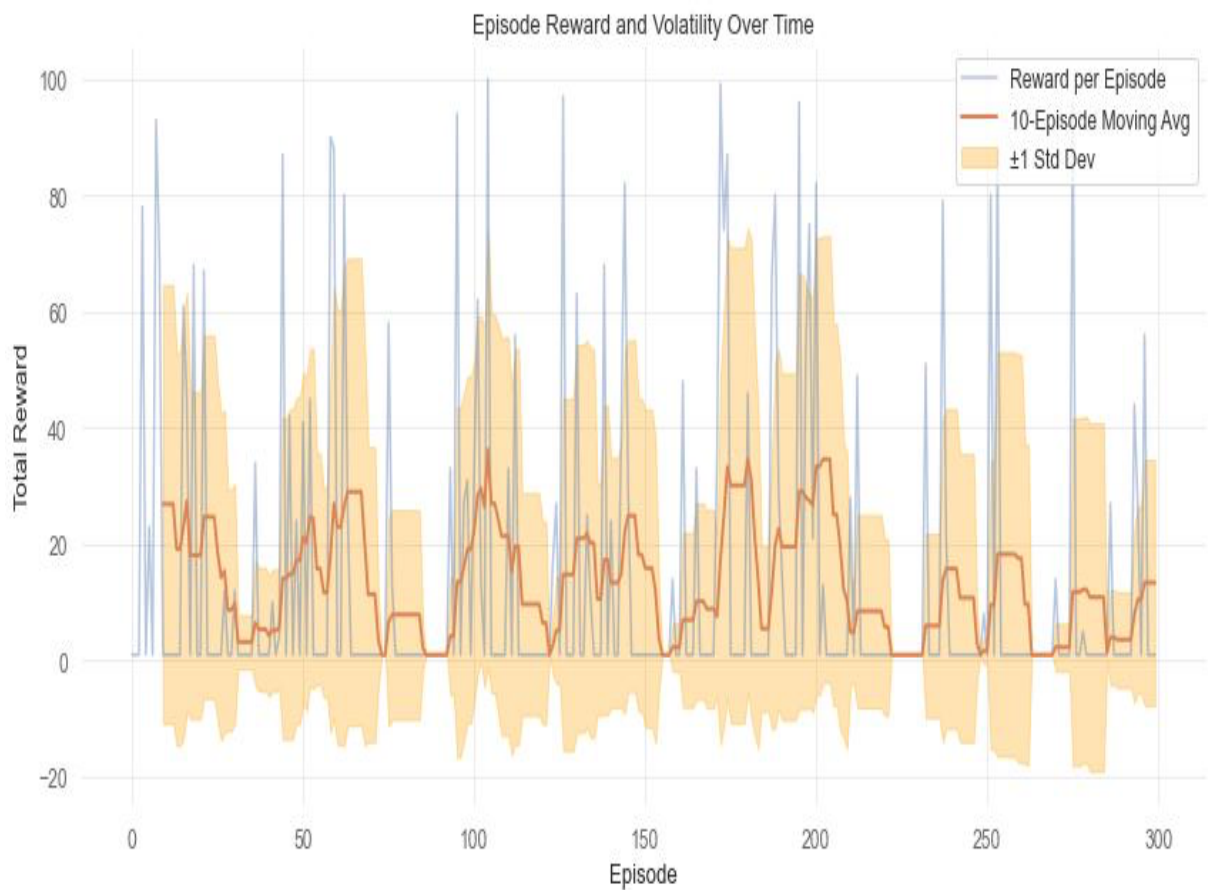
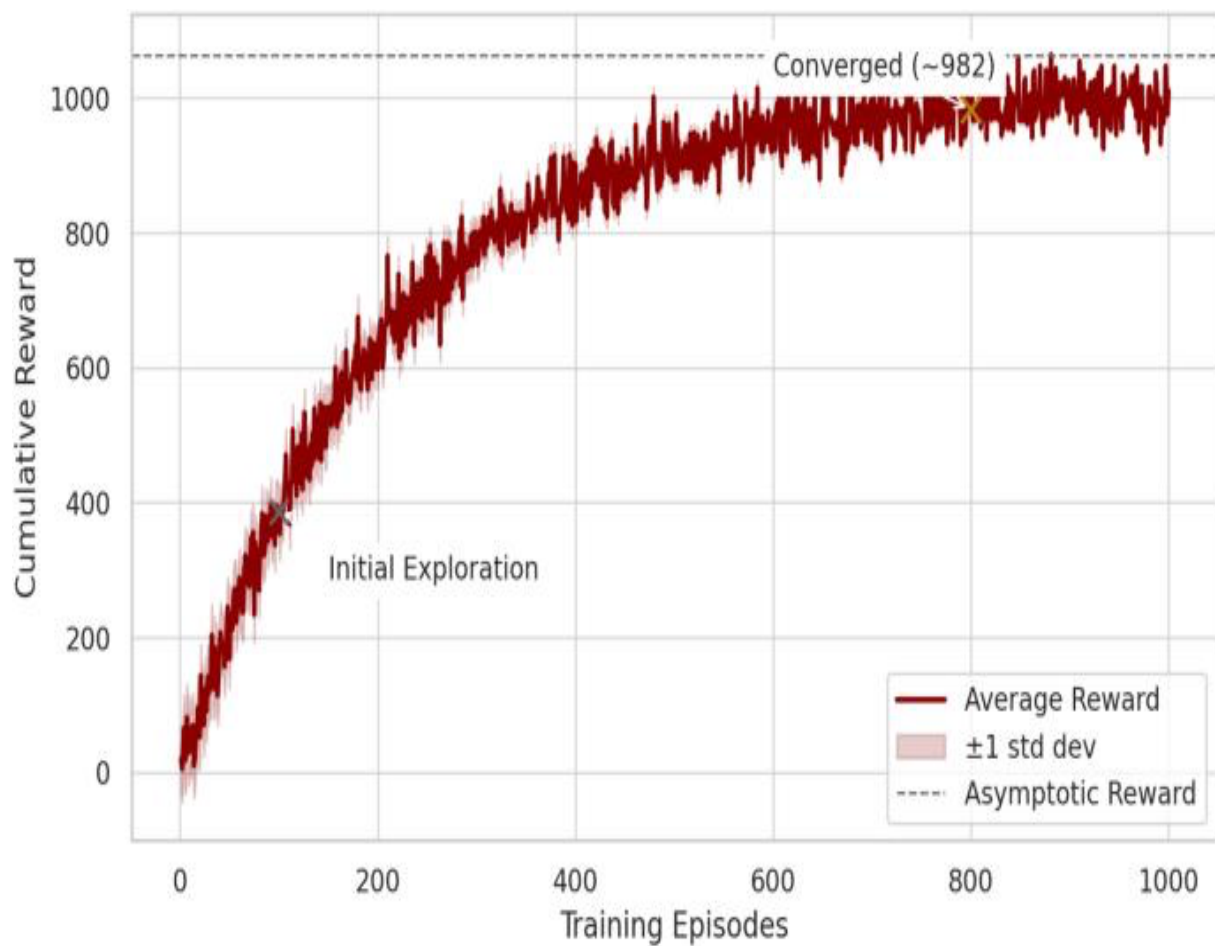
Early training → High fluctuation → Learning → Policy improvement → Stable rewards

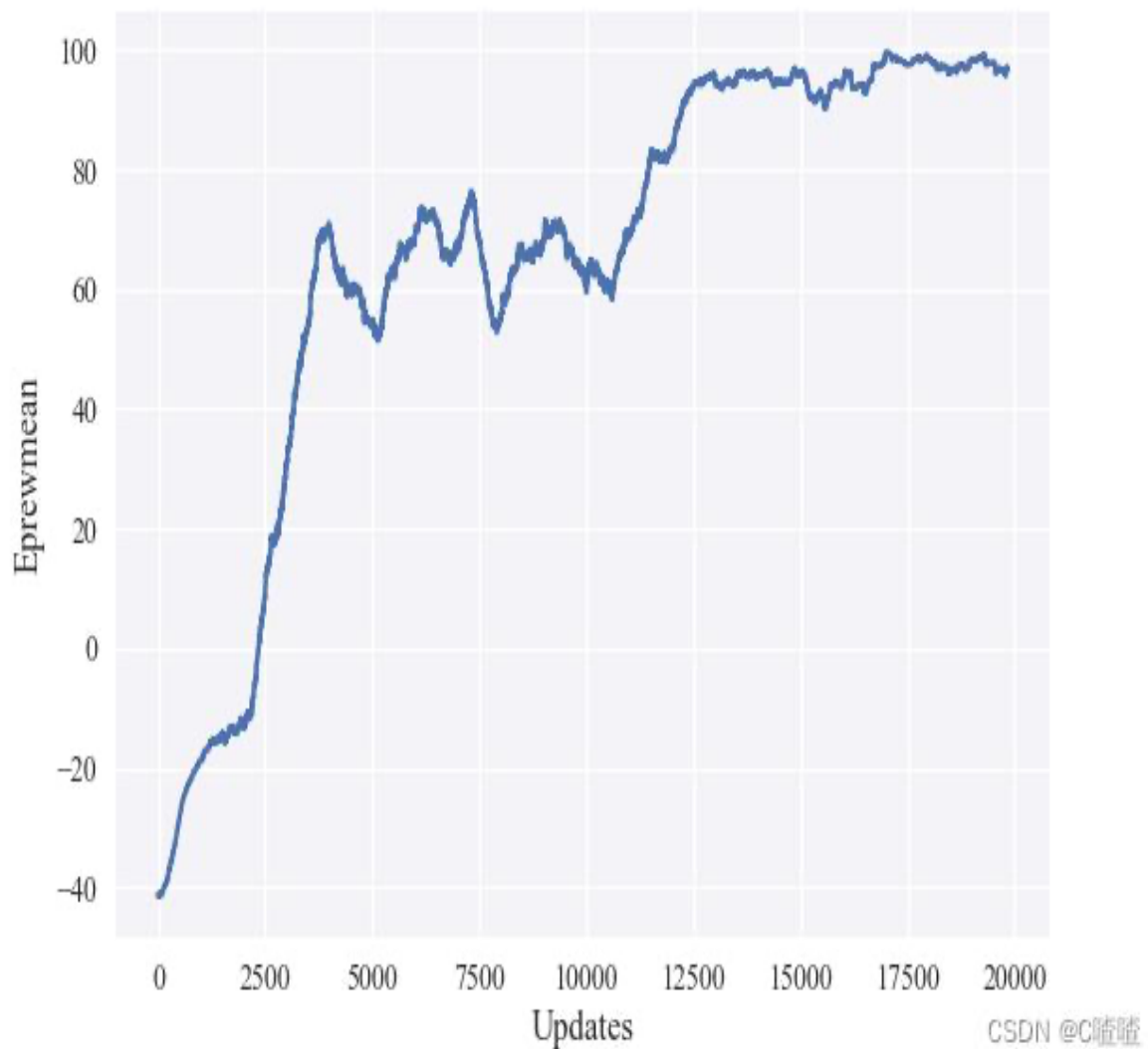
Stable rewards suggest that the agent's decisions are becoming more consistent and that its learned policy is becoming robust.

However, stability alone does not guarantee that the policy is optimal. The stable reward should also be sufficiently high and ideally close to the best achievable performance.

Flow Chart







Conclusion

The fluctuating rewards in the early episodes are normal in a stochastic environment because the agent is exploring and learning. Stabilization after **500 episodes** indicates that the agent has learned a more consistent and robust policy. If the stable rewards are also high, this provides strong evidence of successful learning and convergence.